

Long runs and rare patterns

of a random completely multiplicative function

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Abstract

Let f be a random completely multiplicative function with independent symmetric signs at the primes. At logarithmic run lengths, we approximate the full field of positions, excess lengths and signs by independent Poisson coordinates, despite exact long-range multiplicative identities. More generally, a deterministic dictionary of $m \leq N^{1/2-\eta}$ prescribed words at length $B = \log_2(Nm) + O(1)$ is asymptotically equivalent in total variation to the rare-word field of independent fair signs, when its weighted overlap tends to zero. Most dictionaries satisfy this condition; explicit marker families have no overlaps.

The arithmetic input is a uniform two-window square-relation estimate with critical bound $O_\varepsilon(N^{5/3+\varepsilon})$. A capped form exploits the mutual exclusivity of distinct words at one site and allows the dictionary to grow. The proof uses rational relations, private prime coordinates and bounded-height Runge–Pell arguments. Conditioning on small primes then supplies an exact dependency graph.

For the longest run in $[1, M]$, we obtain the lattice extreme law and asymmetric almost-sure envelopes. Exceptionally long runs arise either from a boundary event of exact mass $2^{-\pi(L)}$ or from a bulk field of intensity $M2^{-L}$. A relative marked comparison gives, conditional on existence, a lattice mixture for the first location, sign and overshoot with moving source weights. The two source branches have geometric overshoots in different clocks: prime rank at the boundary and integer distance in the bulk.

Keywords. Random multiplicative functions; long runs; rare patterns; Poisson approximation; square-product relations; boundary effects.

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1 Introduction

In a sequence of independent fair signs, a long constant run is rare for a simple reason: each additional sign must agree with the preceding one. A left-maximal run of length at least L starts with probability 2^{-L} , so N potential starts have natural intensity $N2^{-L}$. The transition between many runs and no runs occurs at $L = \log_2 N + O(1)$; recording starts rather than all constant windows removes the local clustering within a run [2, 3, 15].

Logarithmic blocks also underlie the original Erdős–Rényi law of large numbers [12]; the distributional and marked run laws used here are the later results cited above.

Now impose complete multiplicativity. Choose the signs $f(p)$ independently at the primes and set

$$f(1) = 1, \quad f(n) = \prod_p f(p)^{v_p(n)}, \quad \mathbb{P}(f(p) = 1) = \mathbb{P}(f(p) = -1) = \frac{1}{2}. \quad (1.1)$$

In particular $f(n^2) = 1$. This is the completely multiplicative, sometimes called *extended Rademacher*, model, not the squarefree-supported model. Random multiplicative models go back to Wintner [36]; the completely multiplicative convention here is fixed by (1.1). The identity $f(2n) = f(2)f(n)$ persists at every scale. Distant blocks therefore cannot simply be treated as independent: the effect of these identities must be counted. Nevertheless, the critical run statistics have the independent-sign Poisson form.

There is a second, quite different mechanism at the left boundary. Requiring $f(1), \dots, f(L)$ to be equal is exactly the requirement that every prime sign up to L be positive. Its probability is $2^{-\pi(L)}$, not 2^{-L} . Thus a sufficiently exceptional run can be produced either by many potential starts in the bulk or by a single arithmetically cheaper event at the boundary. Our results quantify both mechanisms and their competition.

The same approach also distinguishes many prescribed words at once. Two sources of dependence must be separated: words can overlap because a suffix matches a prefix, and distant occurrences can interact because a product of their integer arguments is a square. The first is a property of the dictionary; the second is a property of the arithmetic sequence. Keeping them separate is the organizing principle of the paper.

1.1 Main results

For $x \geq 2$ and $L \geq 1$, let

$$J_{x,L} = \mathbf{1}_{\{f(x-1)=-f(x), f(x)=\dots=f(x+L-1)\}}. \quad (1.2)$$

Write $I_N = [N, 2N) \cap \mathbb{Z}$, $Z_{N,L} = \sum_{x \in I_N} J_{x,L}$ and $\lambda_N = N2^{-L}$. On a start, let ℓ_x be the maximal rightward run length. Every such length is almost surely finite. Define

$$K_{x,e} = \mathbf{1}_{\{J_{x,L}=1, \ell_x=L+e\}}, \quad K_{x,e}^s = K_{x,e} \mathbf{1}_{\{f(x)=s\}}, \quad e \in \mathbb{N}_0, s \in \{\pm 1\}.$$

Let $\nu_{\text{geo}}\{e\} = 2^{-e-1}$ and $\nu_{\pm} = (\delta_{-1} + \delta_{+1})/2$. A *word* is a prescribed finite sign sequence, a *dictionary* is a set of such words, and *pattern* is the broader term for the corresponding occurrence observable. All logarithms without an indicated base are natural. All total-variation distances below use the convention $d_{\text{TV}}(\mu, \nu) = \sup_A |\mu(A) - \nu(A)|$.

Theorem 1.1 (Critical sign-resolved run field). *Fix $C > 0$. Uniformly for integer lengths $|L - \log_2 N| \leq C$,*

$$d_{\text{TV}} \left(\mathcal{L}((K_{x,e}^s)_{x,e,s}), \bigotimes_{x,e,s} \text{Pois}(2^{-L-e-2}) \right) \leq \exp \left[- \left(\frac{1}{\sqrt{2}} + o_C(1) \right) \sqrt{\log N \log \log N} \right]. \quad (1.3)$$

Here (x, e, s) ranges over $I_N \times \mathbb{N}_0 \times \{\pm 1\}$, and the comparison is on the common countable lattice. In particular the same upper bound applies to $d_{\text{TV}}(\mathcal{L}(Z_{N,L}), \text{Pois}(\lambda_N))$.

Since L is integral, changing it by one divides λ_N by two; λ_N need not converge. This is why the approximation retains its moving parameter. For continuous positions, the consequence is a weak limit. Along a subsequence with $\lambda_N \rightarrow \lambda \in (0, \infty)$,

$$\sum_{x \in I_N} \sum_{e \geq 0} \sum_{s \in \{\pm 1\}} K_{x,e}^s \delta_{(x/N, e, s)} \implies \text{PPP}(\lambda dt \otimes \nu_{\text{geo}} \otimes \nu_{\pm})$$

weakly on finite point configurations on $[1, 2] \times \mathbb{N}_0 \times \{\pm 1\}$. The nonempty configurations on a deterministic spatial grid and those of a diffuse Poisson process are singular. Consequently the last convergence is weak, not in total variation. The lattice comparison is stronger in the appropriate prelimit space: a single coupling controls positions, excesses, signs, threshold counts and order statistics simultaneously. In the bulk, positive and negative long runs have equal limiting intensities, despite the deterministic positivity of every square.

Let R_M be the longest constant run in $(f(1), \dots, f(M))$.

Theorem 1.2 (Longest run and competing rare-event sources). *At critical lengths $|L - \log_2 M| \leq C$,*

$$\mathbb{P}(R_M < L) = \exp(-M2^{-L}) + o_C(1). \quad (1.4)$$

Moreover,

$$R_M = \log_2 M + O(\log \log M) \quad \text{almost surely.} \quad (1.5)$$

Suppose instead that $L \rightarrow \infty$, $L = O(\log M)$, and $\lambda_M = M2^{-L} \rightarrow 0$. Then

$$\mathbb{P}(R_M \geq L) = 2^{-\pi(L)} + M2^{-L} + o(2^{-\pi(L)} + M2^{-L}). \quad (1.6)$$

Let $X_{M,L}$ be the leftmost start of a length- L constant window contained in $[1, M]$, on the event $R_M \geq L$. If $L - \log_2 M - \pi(L) \rightarrow s \in \mathbb{R}$, then

$$\mathcal{L}(X_{M,L}/M \mid R_M \geq L) \implies \frac{1}{1 + 2^{-s}} \delta_0 + \frac{2^{-s}}{1 + 2^{-s}} \text{Leb}_{[0,1]}. \quad (1.7)$$

The boundary branch is more precise than its atom at macroscopic position zero: with conditional probability tending to one its start is exactly 1.

The centring in (1.7) compares two exact costs, $L - \log_2 M$ and $\pi(L)$, rather than replacing the prime count by a first-order asymptotic at a precision where that replacement is invalid. Theorem 7.9 strengthens the location law to a lattice comparison with moving source weights, sign and overshoot. Its geometric clock counts integers in the bulk and prime ranks at the boundary; no phase-convergent subsequence is needed for that prelimit comparison.

The third result treats many words at once, with their absolute signs prescribed. Let $\mathcal{W}_N \subset \{\pm 1\}^{B_N}$ be a deterministic set of $m_N \geq 1$ distinct words. Define

$$I_{x,w} = \mathbf{1}_{\{f(x-1+j)=w_j \ (0 \leq j < B_N)\}}, \quad x \in I_N, \ w \in \mathcal{W}_N.$$

Write $w \sim_d v$ if the suffix $(w_d, \dots, w_{B_N-1})$ equals the prefix $(v_0, \dots, v_{B_N-d-1})$, and set

$$\Omega(\mathcal{W}_N) = \frac{1}{m_N} \sum_{w,v \in \mathcal{W}_N} \sum_{d=1}^{B_N-1} 2^{-d} \mathbf{1}_{\{w \sim_d v\}}. \quad (1.8)$$

Theorem 1.3 (Growing dictionaries of rare words). *Fix $C > 0$ and $0 < \eta < 1/2$. Suppose*

$$m_N \leq N^{1/2-\eta}, \quad |B_N - \log_2(Nm_N)| \leq C, \quad \Omega(\mathcal{W}_N) \rightarrow 0.$$

Then

$$d_{\text{TV}} \left(\mathcal{L}((I_{x,w})_{x \in I_N, w \in \mathcal{W}_N}), \bigotimes_{x \in I_N, w \in \mathcal{W}_N} \text{Pois}(2^{-B_N}) \right) \rightarrow 0. \quad (1.9)$$

The same conclusion holds after any deterministic restriction of the site-word index set. The rare-word field is also asymptotically equivalent in total variation to the field formed from independent fair signs.

The total target intensity is $Nm_N 2^{-B_N}$, while each word has mean $N 2^{-B_N}$. Thus the comparison retains both the location and identity of a rare word, even when the dictionary grows. The theorem is proved quantitatively in Theorem 5.1; Corollary 5.3 shows that all but an $O(\log N / \sqrt{N})$ fraction of dictionaries of the stated size have a suitably small overlap weight. There are also explicit deterministic examples with no overlaps at all: Corollary 5.4 uses a classical binary marker construction. For example, $\lfloor N^{2/5} \rfloor$ prescribed words of length $(7/5) \log_2 N + O(1)$ can be treated simultaneously. In this example the number of word types grows, but the expected total number of occurrences stays bounded. These are different parameters, and the theorem controls the former without discarding any labels.

For a single word counted up to sign, take $\mathcal{W} = \{a, -a\}$. The run-start word $a = (-1, +1, \dots, +1)$ has overlap weight 2^{-L} . Dictionaries with nonvanishing weighted overlap are outside the theorem; short-period words illustrate the possible local clusters. The compound-Poisson law in Corollary 5.7 describes the clusters produced by counting every constant window rather than only run starts.

The three main results use the same comparison in three forms: the raw profile controls run fields, the capped profile handles growing families of mutually exclusive words, and the relative comparison resolves the competition between the exact boundary event and the bulk.

1.2 The mechanism and the arithmetic estimate

Write prime signs as $f(p) = (-1)^{X_p}$ with independent uniform $X_p \in \mathbb{F}_2$. A pattern up to sign prescribes L affine linear conditions on these prime coordinates. For two separated windows, let $\rho_L(x, y)$ be the dimension of the homogeneous relation space of their stacked systems. It depends on the windows, not on the prescribed signs. The affine Fourier identity gives

$$\left| \mathbb{P}(I_x = I_y = 1) - 2^{-2L} \right| \leq 2^{-2L} (2^{\rho_L(x,y)} - 1).$$

Thus the useful arithmetic object counts *all nonzero relations*, with their full multiplicity, rather than merely the pairs that carry one. For prescribed absolute words the relevant kernel omits the two block-parity conditions. Its dimension increases by at most two. A host count valid without parity restrictions shows that the same weighted bound survives; this is the first bridge from relative signs to the dictionary theorem.

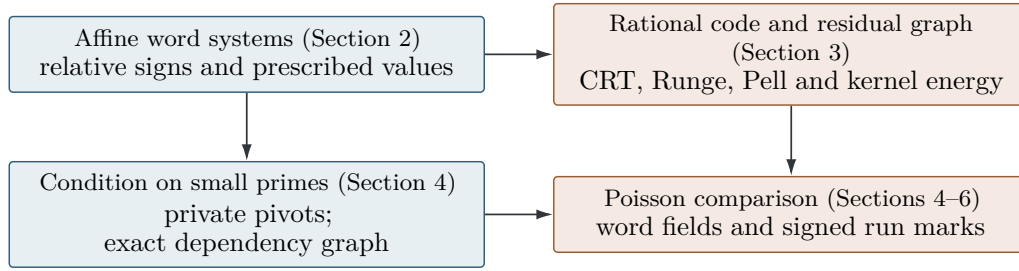


Figure 1: The arithmetic and probabilistic interfaces. The weighted two-window estimate bounds the separated-pair term. Private pivots supply the exact conditional graph and the correct marginals; local overlap is treated separately. Neither interface is dispensable.

The second bridge is probabilistic: at a fixed site the dictionary words are mutually exclusive, so the chance of a joint occurrence cannot exceed a single-site probability. Capping the relation weight at this ceiling, before summing over pairs, is what permits dictionaries of size $N^{1/2-\eta}$.

The uniform estimate proved in section 3 is, for fixed $0 < \delta < 1$ and $B = L + 1 \asymp \log M$,

$$\sum_{\substack{M^\delta \leq x, y < M \\ |x-y| > L}} (2^{\rho_L(x,y)} - 1) \ll_\varepsilon M^\varepsilon \left(M^{3/2} 2^{B/6} + M 2^{2B/3} + M^{2/3} 2^B \right). \quad (1.10)$$

The fixed logarithmic band and δ are understood in the implicit constant. This single geometry includes the dyadic and bounded-ratio estimates by restriction. At criticality the bound is $M^{5/3+\varepsilon}$.

Two devices make the proof work. Exact rational coincidences are first removed as a systematic binary code. The remaining relations are resolved by the graph of prime coordinates larger than B . For the hardest terminal population, determinants force many small odd kernels into the same short window. Counting the kernels separately is wasteful; a shifted pair-energy estimate detects the necessary concentration. Finally one conditions on prime signs below a larger cutoff Y . Most starts then have exact conditional marginal 2^{-L} , and disjoint prime supports give an exact conditional dependency graph. Chen–Stein is applied *after* this arithmetic separation, not as a substitute for it.

1.3 Relation to earlier work

The independent-sign theory supplies the run-start normalization, the geometric excess law and the lattice extreme-value comparison [14, 17, 3]. Dependency-graph Poisson approximation is classical [1, 2, 9]. Poisson approximation for occurrences of prescribed words and the role of self-overlap are also classical [15, 16]. For several words, Reinert–Schbath [28] give Poisson-process and compound-Poisson approximations for clumps in Markov chains; Roquain–Schbath [30] treat overlapping rare word families. Thus a joint approximation with word types is not itself a new probabilistic principle. The contribution here is to verify joint, site-labelled comparisons in a nonstationary arithmetic sequence with exact multiplicative identities, uniformly for logarithmic word length and a growing dictionary.

Najnudel proves eventual independence of fixed-length blocks and almost-sure equidistribution of fixed words in roots-of-unity models [25]. In the sign model the individual laws can be degenerate at squares, so this independence does not say that every coordinate is a fair sign. Fixed-pattern and sign-complexity questions also arise for deterministic multiplicative functions [6, 23, 22]. Our growing block length and simultaneous count over moving starts require uniform

weighted control of pairs of windows, beyond a fixed-block independence statement. The pattern theorem makes explicit which part of that control is insensitive to the affine right-hand side.

Square-product paucity and Diophantine rigidity are important tools in random multiplicative functions. Related applications include short-interval central limit theorems and random Chowla problems [7, 32, 19, 10]. For fixed polynomial values, de la Bretèche, Wang and Xu use square-product counts and Evertse–Silverman or Pell estimates [11]. Munsch–Shparlinski–Sun–Xiao [24, Theorem 1.3] prove an unconditional square-product bound, uniform in the translation of one interval, for each fixed number of factors. The constants depend on that fixed number; the result does not directly control all relation weights in two moving windows of logarithmic length. Tao’s study of products of consecutive integers with unusual anatomy provides a particularly close comparison for the smooth-number and Diophantine mechanisms [33]. These works do not provide the growing-window profile (1.10) as an immediate specialization. The properly normalized short-interval sums of Harper, Soundararajan and Xu [18] concern a different observable and, in their stated model, Steinhaus prime values rather than binary run indicators.

We use Runge’s method in an explicitly proved split-product form [31, 35]; for fixed-degree component counts, bounded-height square-class localization and a Pell equation suffice (Lemma 3.12). Uniform integral-point theorems such as Evertse–Silverman [13] provide stronger, height-free information, but are not inputs to our proof. The microscopic analysis uses the uniform prime-divisor lower bound of Laishram–Shorey [21] and the pointwise result of Balasubramanian–Shorey [4]. The geometric marked target, its thinning chain and its Gaussian limits are probabilistic consequences once the arithmetic comparison is established; they are not presented as new properties of Poisson processes.

1.4 Reading guide and scope

Section 2 develops the affine encoding and the one-window bounds. Section 3 proves the final uniform two-window estimate and its extension to prescribed values. Sections 4–6 extract scalar, pattern, marked and conditional consequences. Section 7 handles the finite prefix, the boundary and the relative crossover. A reader primarily interested in probability may take the uniform, prescribed-value and capped two-window profiles as arithmetic inputs (Theorem 3.1 and propositions 3.24 and 3.25), then continue with Sections 4–7. The two-window proof includes its Diophantine counting arguments, using standard ideal and unit theory for the Pell step. Section 7 adds the explicitly stated prime-divisor and smooth-product inputs needed near the boundary. The companion expands the technical steps called from the article, rather than giving a second proof of the relation estimate. The exact prime-cutoff definitions in Section 4 are not needed to read the three main statements.

We do not assert optimality of the exponent $5/3$ or of the prime-cutoff ranges. The full one-factor range for the spatially labelled field remains unproved. The dictionary-size exponent is likewise a sufficient bound, not an optimality statement. Records, occupation laws, scale flows and auxiliary phase removal are developed in the separate process article [27]; none is a proof input here.

2 Affine events and private prime coordinates

A run event is a linear system over $\mathbb{F}_2$. Its probability is determined by rank and compatibility; its dependence on a second event is determined by relations between the two systems. This elementary observation lets us keep arithmetic and probability separate. For $n \geq 1$, let

$$v(n) = (v_p(n) \bmod 2)_p \in \mathbb{F}_2^{(\mathcal{P})}.$$

Write the prime signs additively: $X = (X_p)_{p \in \mathcal{P}}$ denotes the sequence of independent Bernoulli variables, uniform on $\mathbb{F}_2$, defined by $f(p) = (-1)^{X_p}$. Then

$$f(n) = (-1)^{\langle X, v(n) \rangle}.$$

For a start x , introduce the L rows

$$\begin{aligned} a_{x,0} &= v(x-1) + v(x), & b_{x,0} &= 1, \\ a_{x,j} &= v(x) + v(x+j), & b_{x,j} &= 0, \quad 1 \leq j < L. \end{aligned}$$

Let A_x be the matrix with rows $a_{x,j}$ and $b_x = (1, 0, \dots, 0)$. Then

$$J_{x,L} = \mathbf{1}_{\{A_x X = b_x\}}.$$

For a tuple $\mathbf{x} = (x_1, \dots, x_k)$, stack the matrices and the right-hand sides. Set

$$\mathcal{R}(\mathbf{x}) = \ker(A_{\mathbf{x}}^\top), \quad \rho(\mathbf{x}) = \dim \mathcal{R}(\mathbf{x}),$$

and

$$q_{\mathbf{x}}(c) = \langle c, b_{\mathbf{x}} \rangle.$$

Lemma 2.1 (Affine Fourier identity). *For every tuple $\mathbf{x} = (x_1, \dots, x_k)$,*

$$2^{kL} \mathbb{P}(J_{x_1,L} = \dots = J_{x_k,L} = 1) = \sum_{c \in \mathcal{R}(\mathbf{x})} (-1)^{q_{\mathbf{x}}(c)}.$$

Consequently,

$$\mathbb{P}(J_{x_1,L} = \dots = J_{x_k,L} = 1) = 2^{-kL} \eta(\mathbf{x}) 2^{\rho(\mathbf{x})},$$

where $\eta(\mathbf{x}) = 1$ if $q_{\mathbf{x}}$ vanishes on $\mathcal{R}(\mathbf{x})$, and 0 otherwise. Finally,

$$\left| \eta(\mathbf{x}) 2^{\rho(\mathbf{x})} - 1 \right| \leq 2^{\rho(\mathbf{x})} - 1. \quad (2.1)$$

Proof. For a matrix with $m = kL$ rows,

$$\mathbf{1}_{\{AX=b\}} = 2^{-m} \sum_{c \in \mathbb{F}_2^m} (-1)^{\langle c, AX-b \rangle}.$$

Taking expectations, the terms with $A^\top c \neq 0$ vanish. A character sum over a vector space equals the cardinality of that space if the character is trivial, and 0 otherwise. If $\eta = 0$, then q is nontrivial, hence $\rho \geq 1$, and $1 \leq 2^\rho - 1$. $\square$

2.1 The boundary map of a tree

Put $B = L + 1$, $I_L = \{-1, 0, \dots, L-1\}$ and $V_x = \{x + i : i \in I_L\}$. The rows of A_x are the edge vectors of the star with edges $\{x-1, x\}$ and $\{x, x+j\}$, $1 \leq j < L$. For a finite tree, the map taking an edge set to the vertices of odd degree is a bijection onto the even vertex subsets. Thus a relation among the rows of A_x selects an even subset of V_x whose product is a square.

Lemma 2.2 (Square-product interpretation). *For disjoint windows $V_{x_1}, \dots, V_{x_k}$, the relation space is in bijection with tuples of even subsets $S_i \subset V_{x_i}$ such that $\prod_i \prod_{n \in S_i} n$ is a square. The affine character is $\sum_i \mathbf{1}_{\{x_i-1 \in S_i\}}$ modulo two. More generally, the edge relations of any tree are identified with its square-product even vertex subsets.*

Proof. The sum of the selected edge vectors is the sum of the valuation vectors at the odd-degree vertices. It vanishes exactly when their product is a square. The boundary map is injective because a tree has no nonempty even degree edge subgraph, and its image has dimension $|V| - 1$, the dimension of the even-subset space. The affine character records the root leaf edges. $\square$

2.2 One-window arithmetic

For $H \geq 2$ define the odd large-prime kernel

$$K_H(n) = \prod_{\substack{p > H \\ v_p(n) \text{ odd}}} p.$$

Call n H -defective when $K_H(n) = 1$. If an interval has diameter less than H , every nondefective vertex has a prime coordinate appearing there and nowhere else in the interval. Such a coordinate is a private pivot. Notice that the definition uses odd valuations, not merely the presence of a large prime divisor.

Lemma 2.3 (Split-product Runge bound). *There is an absolute C_0 such that the following holds. Let $d = 2k \geq 2$, let $\gamma_1, \dots, \gamma_d$ be distinct integers with $|\gamma_i| \leq R$, where $R \geq 1$, and set $G(T) = \prod_i (T + \gamma_i)$. If $U \in \mathbb{Z}$, $U \geq 2R$, and $G(U)$ is an integer square, then*

$$U \leq (C_0 d R)^{2d}. \quad (2.2)$$

Proof. Expand $\sqrt{G(T)}$ at infinity and let $P(T)$ be its polynomial part. Its coefficients lie in $2^{-d}\mathbb{Z}$ and, for $U > 4R$, $|\sqrt{G(U)} - P(U)| \leq (CdR)^d/U$. If the difference is nonzero, integrality of U and $\sqrt{G(U)}$ makes its modulus at least 2^{-d} . Otherwise the nonzero integer polynomial $2^{2d}(G - P^2)$ vanishes at U ; its coefficient height is at most $(CdR)^{2d}$, and the integral-root bound applies. The range $2R \leq U \leq 4R$ is immediate. All constants and the growing-degree coefficient estimates are given in Lemma A.1. This is a direct split-product implementation of Runge's method, not an unqualified invocation of a fixed-degree theorem [31, 35]. $\square$

Proposition 2.4 (Few defects, with their exponential weight). *Fix $0 < c_1 < c_2$. Uniformly for integers $c_1 \log X \leq H \leq c_2 \log X$, an interval of $H + 1$ integers in $[X/2, 3X]$ contains $O_{c_1, c_2}(\log X / \log \log X)$ H -defective integers. If m_u counts these defects in $[u, u + H]$, then*

$$\sum_{X \leq u < 2X} (2^{m_u} - 1) \leq X^{1/2 + o_{c_1, c_2}(1)}. \quad (2.3)$$

Proof. Write each defective integer as sa^2 , where s is squarefree and H -smooth. Associate to the m defects in a window the columns $((v_p(s))_{p \leq H}, 1) \in \mathbb{F}_2^{\pi(H)+1}$. A nonzero kernel word selects an even number d of distinct integers with square product. The Runge bound gives $X/2 \leq (CdH)^{2d}$; consequently its Hamming weight is at least $c \log X / \log H$ for a fixed $c > 0$. Let t be an integer of order $\log X / \log H$ with $2t + 1 \leq c \log X / \log H$. If the kernel is zero, $m \leq \pi(H) + 1$. If $m < 2t$, the desired bound is again immediate. Otherwise disjoint Hamming balls about its words give

$$\sum_{j \leq t} \binom{m}{j} \leq 2^{\pi(H)+1}, \quad (m/t)^t \leq 2^{\pi(H)+1}.$$

Since $(\pi(H) + 1)/t = O_{c_1, c_2}(1)$, it follows that $m = O_{c_1, c_2}(\log X / \log \log X)$.

On the other hand the number of defective integers up to $3X$ is at most

$$\sqrt{3X} \sum_{\substack{s \text{ squarefree} \\ P^+(s) \leq H}} s^{-1/2} = \sqrt{3X} \prod_{p \leq H} (1 + p^{-1/2}) = X^{1/2+o(1)}.$$

Each belongs to at most $H + 1 = X^{o(1)}$ windows. Only $X^{1/2+o(1)}$ windows therefore have a defect, and the pointwise bound makes each weight 2^{m_u} equal to $X^{o(1)}$. This proves the sum. $\square$

For $0 < \delta < 1$, write $U_{M,\delta} = [\lceil M^\delta \rceil, M) \cap \mathbb{Z}$. If $B \asymp \log M$, a dyadic slice $X \leq x < 2X$ in this interval has $B \asymp_\delta \log X$. Applying proposition 2.4 on these slices gives

$$\sum_{x \in U_{M,\delta}} (2^{m_B(x)} - 1) \leq M^{1/2+o(1)}, \quad m_B(x) = \#\{n \in V_x : K_B(n) = 1\}. \quad (2.4)$$

The same assertion holds for windows whose width and defect threshold are any two comparable quantities in a fixed logarithmic band: use an enclosing window of width equal to a fixed multiple of the threshold and subdivide it into a bounded number of windows before the coding argument. In the applications below the threshold can always be chosen at least the width.

Corollary 2.5 (Uniform first moment and an affine bound). *Put $p = 2^{-L}$ and $m_B^\circ(x) = \#\{n \in V_x \setminus \{x-1\} : K_B(n) = 1\}$. For any right-hand side $b \in \mathbb{F}_2^L$,*

$$|\mathbb{P}(A_x X = b) - p| \leq p(2^{m_B^\circ(x)} - 1), \quad \mathbb{P}(A_x X = b) \leq 2^{-L + \max(m_B(x) - 1, 0)}. \quad (2.5)$$

For every deterministic $A \subset I_N$ and B in a fixed logarithmic band,

$$\mathbb{E} \sum_{x \in A} J_{x,L} = |A|p + O(pN^{1/2+o(1)}).$$

The analogous error over any subset of $U_{M,\delta}$ is $O(pM^{1/2+o(1)})$.

Proof. A private pivot excludes its vertex from any relation. An even subset of m defective vertices has at most $\max(m-1, 0)$ degrees of freedom. Thus $\rho(x) \leq \max(m_B(x) - 1, 0) \leq m_B^\circ(x)$. Apply Lemma 2.1, which is unchanged by the affine right-hand side, and sum proposition 2.4 or (2.4). Restricting a nonnegative error sum to a mask can only decrease it. $\square$

For $Y \geq 2$ write $\mathcal{F}_Y = \sigma(f(p) : p \text{ prime}, p \leq Y)$ for the information carried by the small prime signs.

Corollary 2.6 (Prescribed values in one window). *Let M_x have rows $v(n)$, $n \in V_x$, and let $w \in \{\pm 1\}^B$. Write $I_{x,w} = \mathbf{1}_{\{(f(x-1+j))_{0 \leq j < B} = w\}}$ and $p_B = 2^{-B}$. Then*

$$|\mathbb{P}(I_{x,w} = 1) - p_B| \leq p_B(2^{m_B(x)} - 1). \quad (2.6)$$

If, for every vertex $n \in V_x$, there is a prime $p > Y$, where $Y > B$, such that $v_p(n)$ is odd, then $\mathbb{P}(I_{x,w} = 1 \mid \mathcal{F}_Y) = p_B$ for every realization of $\mathcal{F}_Y$. Fix $0 < c_1 < c_2$. Uniformly for $c_1 \log N \leq B \leq c_2 \log N$, every deterministic start mask $A \subset I_N$ and dictionary $\mathcal{W} \subset \{\pm 1\}^B$ satisfy

$$\mathbb{E} \sum_{x \in A} \sum_{w \in \mathcal{W}} I_{x,w} = |A| |\mathcal{W}| 2^{-B} + O_{c_1, c_2}(|\mathcal{W}| 2^{-B} N^{1/2+o_{c_1, c_2}(1)}).$$

The expectation is unconditional. The threshold and error are uniform in $B, A, \mathcal{W}$, which may vary with N ; empty masks and dictionaries are allowed. The mask restricts starts, not vertices:

every occurrence uses all B values $x - 1, \dots, x + B - 2$, without truncation at the mask or at the boundary of I_N .

Proof. For a fixed window of positive integers, take a prime cutoff at least as large as its largest vertex, and at least Y when conditioning. The word event in the infinite product is the inverse image of the corresponding finite-cylinder event, so their probabilities agree exactly. Only the prime signs are independent; the integer values retain all their multiplicative relations. In a relation among the rows of M_x , every nondefective vertex is excluded by its private coordinate. Hence the nullity is at most $m_B(x)$, without the even-subset restriction used for relative signs. Finite Fourier inversion gives (2.6). On a good support, the projection of M_x onto primes above Y has an identity minor of order B , so every prescribed value vector has probability 2^{-B} on each positive-mass assignment atom of $\mathcal{F}_Y$. Sum the nonnegative bound using proposition 2.4, then use linearity of expectation. The same threshold works for every mask and every dictionary; no independence between occurrences is needed. $\square$

Lemma 2.7 (Private pivots on a graph). *Let G be a finite graph. In each connected component suppose all vertices except possibly one have distinct private prime coordinates in the whole graph. The kernel of its edge vectors, projected onto these coordinates, is contained in its cycle space. In particular its rank is at least $|E(G)| - \beta_1(G)$, where $\beta_1(G) = |E(G)| - |V(G)| + c(G)$. For a tree this rank is $|E(G)|$.*

Proof. A vanishing edge combination has even degree at every pivoted vertex. The sum of degrees in a connected component is even, so its possible unpivoted vertex has even degree as well. The selected edge set is therefore in the cycle space, of dimension $\beta_1(G)$. $\square$

Lemma 2.8 (Local run pairs). *If $0 < |x - y| < L$, then $J_{x,L}J_{y,L} = 0$. Over ordered pairs with $|x - y| = L$ in a dyadic block or in $U_{M,\delta}$, the homogeneous weights $2^{\rho(x,y)}$ have sum at most $N^{1+o(1)}$ or $M^{1+o(1)}$, respectively.*

Proof. An overlapping start requires two consecutive signs within the first constant run to be opposite. At distance L , the two start trees form a tree on an interval of diameter $2L$. With threshold $2L + 1$, private pivots and proposition 2.4 bound its relation dimension by $O(\log N / \log \log N)$ on a slice. There are $O(N)$ such pairs; summing slices gives the macroscopic assertion. $\square$

3 A uniform two-window relation estimate

The transfer in Theorem 4.1 separates four costs: weighted one-window defects, supports removed at the prime cutoff, shared-prime graph edges, and the weighted relations estimated here. Only the last requires the two-window arithmetic. We sum $2^{\rho(x,y)} - 1$, not merely count pairs with a relation, by separating exact rational identities from residual prime relations in one macroscopic geometry. Word overlaps are treated separately in Theorem 5.1.

Fix $0 < \delta < 1$ and $0 < \beta_- < \beta_+ < \infty$. Throughout this section,

$$U = U_{M,\delta}, \quad \beta_- \log M \leq B = L + 1 \leq \beta_+ \log M, \quad Q_B = 2^B. \quad (3.1)$$

A factor $M^{o(1)}$ is uniform in this band, with dependence only on the fixed parameters (and on an explicitly fixed component size when one is used). All these parameters are fixed before M . Every occurrence lies in

$$M^\delta - 1 \leq n \leq M + B \leq 2M. \quad (3.2)$$

Theorem 3.1 (Uniform weighted relation profile). *For every $\varepsilon > 0$,*

$$R_{2,\delta}(M, L) := \sum_{\substack{x, y \in U \\ |x-y| > L}} (2^{\rho_L(x, y)} - 1) \\ \ll_{\delta, \beta_-, \beta_+, \varepsilon} M^\varepsilon \left(M^{3/2} Q_B^{1/6} + M Q_B^{2/3} + M^{2/3} Q_B \right) \quad (3.3)$$

$$\ll_{\delta, \beta_-, \beta_+, \varepsilon} M^\varepsilon \left(M^{5/3} + M^{2/3} Q_B \right). \quad (3.4)$$

The defining sum is over ordered pairs. In particular $R_2(N, L) \ll_{C, \varepsilon} N^{5/3+\varepsilon}$ when $|L - \log_2 N| \leq C$ in a dyadic block. The same conclusion, with a constant depending on the fixed upper ratio, holds on $[N, \kappa N]$.

The last assertions follow by restriction from (3.4) at $M = 2N$ or $M = \kappa N$ and, for example, $\delta = 1/2$; the lower endpoint is then at least M^δ for large N . No sum of independent block laws is being taken. The proof of the estimates occupies the rest of this section.

There are three obstacles to counting the relations. Exact rational identities may create many of them for one identifiable reason; we remove that subspace first. Residual relations are then controlled by combining rarity of their hosts with a bound on their dimension. In the terminal case neither resource is sufficient separately: determinants force several small kernels into one window, and it is this simultaneous concentration, rather than the size of the candidate set, that must be counted. The partition below distinguishes which of these resources is available.

3.1 Rational identities and the residual graph

For a separated pair, use the even-subset description in Lemma 2.2. An exact rational relation can be present before any primes are exposed. For example, with $x = 100$, $y = 67$ and $L = 3$, the windows contain the two units $(99, 66)$ and $(102, 68)$ of ratio $3 : 2$. Their product is 6732^2 . Each selected subset in a window has two vertices, so this gives a nonzero homogeneous relation; its two root occurrences also make the affine character even. In fact the individual ranks are three and the stacked rank is five, giving joint probability $1/32$, rather than $1/64$ for independent starts.

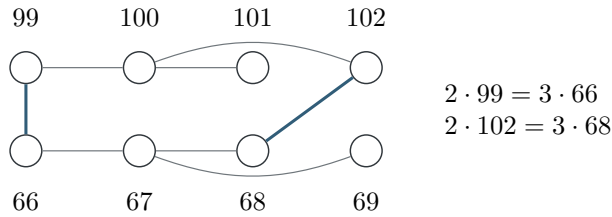


Figure 2: An exact two-unit relation. The thick edges are rational units; the thin edges encode the affine start trees. Even subsets of the units produce relations. This macroscopic example also illustrates why counting all prime coincidences without quotienting the rational subspace loses information.

For coprime $a, b \geq 1$, set $h = by - ax$ and

$$\mathcal{M}_{a,b}(h) = \{(i, j) \in I_L^2 : ai - bj = h\}, \quad m(a, b, h) = |\mathcal{M}_{a,b}(h)|.$$

A member of $\mathcal{M}_{a,b}(h)$ is an exact unit, because $a(x + i) = b(y + j)$. Even subsets of the exact units form a subspace $\mathcal{S}_{a,b,h} \subset \mathcal{R}(x, y)$.

Lemma 3.2 (Rational code and height). *If $m = m(a, b, h) \geq 1$, then $\dim \mathcal{S}_{a,b,h} = m - 1$. If $m \geq 2$, then*

$$\max(a, b) \leq L < B.$$

This conclusion is independent of the ratio between the endpoints x and y .

Proof. For an exact unit, coprimality gives integers t with $x + i = bt$ and $y + j = at$. An even collection E of units satisfies

$$\prod_{(i,j) \in E} (x + i)(y + j) = (ab)^{|E|} \left(\prod_{(i,j) \in E} t \right)^2,$$

which is a square. The even subsets of an m -element set have dimension $m - 1$. Distinct solutions of $ai - bj = h$ differ by (b, a) ; since both offsets remain in an interval of diameter L , one has $a, b \leq L$. $\square$

Fix once and for all $A = 3$. Call (x, y) aligned if there are coprime $a, b \leq B^A$ such that

$$|by - ax| < (a + b)B. \quad (3.5)$$

Lemma 3.3 (Uniqueness of the macroscopic channel). *For M sufficiently large in terms of $(\delta, \beta_-, \beta_+)$, an aligned pair has at most one reduced fraction (a, b) satisfying (3.5). Every channel with at least two exact units is aligned and therefore has this unique fraction.*

Proof. If (a, b) and (a', b') both satisfy (3.5), then

$$|(a'b - ab')x| < 4B^{2A+1}.$$

But $x \geq M^\delta$ and $B = O(\log M)$, so the right-hand side is $o(M^\delta)$. The integer $a'b - ab'$ is therefore zero for large M , and reducedness gives $(a, b) = (a', b')$. If a channel has two units, Lemma 3.2 gives $a, b \leq L$, and for any unit $h = ai - bj$ satisfies $|h| < (a + b)B$. $\square$

When the canonical aligned channel has $m \geq 2$, define

$$\mathcal{S}_{\text{rat}} = \mathcal{S}_{a,b,h}, \quad \sigma = m - 1.$$

Otherwise define $\mathcal{S}_{\text{rat}} = 0$ and $\sigma = 0$. In all cases put

$$\tau = \rho - \sigma.$$

Thus no contraction is made when $m_{\text{ex}} \leq 1$.

Remark 3.4 (Disparate scales). Suppose, for example, that $x \asymp M^\delta$ and $y \asymp M$. If there were two exact units, then $a, b \leq L$ and $x + i = bt$, $y + j = at$, so

$$y \leq L(x + B) + B = o(M),$$

a contradiction. Hence every genuinely disparate-scale pair has $m_{\text{ex}} \leq 1$. It remains in the ordinary large-prime graph and is treated by the small-product, shallow-core or deep-core bounds below; no illegitimate valuation identity or code contraction is used.

Now form the graph $G_{>B}(x, y)$ on the occurrence set $\mathcal{O} = V_x \sqcup V_y$. For a prime $p > B$, let $\mathcal{O}_p$ be the occurrences at which v_p is odd. Each block has diameter $B - 1 < p$, so $|\mathcal{O}_p| \leq 2$, with at most one occurrence in each block. If $|\mathcal{O}_p| = 1$, pin that vertex to zero; if $|\mathcal{O}_p| = 2$, join the

two vertices; parallel equations are deduplicated. A vertex with no large-prime label is defective. Let D be the number of defective vertices and c the number of nontrivial unpinned components.

Lemma 3.5 (Resolution and square class of components). *The solution space of the $p > B$ equations is canonically isomorphic, after one representative is chosen in each component, to*

$$\mathbb{F}_2^D \oplus \mathbb{F}_2^c.$$

In particular, $D + 2c \leq 2B$. Every nontrivial unpinned component C meets both blocks and has a unique representation

$$\prod_{n \in C} n = d_C z_C^2,$$

where d_C is positive, squarefree, and B -smooth.

Proof. A pinned component is zero. All variables in an unpinned component containing an edge are equal and contribute one free coordinate. An isolated defective vertex contributes one free coordinate. Every nontrivial component uses at least two occurrences, giving $D + 2c \leq 2B$.

Every edge is bipartite. For each $p > B$, an unpinned component contains either both odd occurrences of p or none; a single odd occurrence would pin it. Hence the total p -valuation of the product of the component is even for every $p > B$. Its squarefree part is therefore supported on $p \leq B$, and uniqueness is immediate. $\square$

Assume $m \geq 2$. An exact unit has the form $x + i = bt$, $y + j = at$ with $a, b < B$. For every $p > B$, $p \nmid ab$ and therefore

$$v_p(x + i) \equiv v_p(t) \equiv v_p(y + j) \pmod{2}.$$

If $K_B(t) = 1$, the two occurrences are defective; otherwise they form an isolated unpinned component of size two. Distinct exact units use disjoint occurrences. Let d be the number of exact units of the first type and $n = m - d$ the number of the second type, and define

$$D^\# = D - d, \quad c^\# = c - n,$$

removing the n exact components from the residual component set $\mathcal{C}_{\text{res}}$. If $m < 2$, set $D^\# = D$, $c^\# = c$, and let $\mathcal{C}_{\text{res}}$ be all nontrivial unpinned components. Notice that $D^\#$ is a dimensional correction, not literally a number of remaining defective vertices.

Lemma 3.6 (Quotient core). *Uniformly for all separated pairs,*

$$\tau \leq D^\# + c^\#, \quad D^\# + 2c^\# \leq 2B, \quad D^\# = O(B/\log B).$$

Proof. For $m \geq 2$, the m exact-unit vectors are independent in the large-prime solution space and the even code has dimension $m - 1$. The parity functional of the first block descends to the quotient and is nonzero there; hence the even-first-block subspace has dimension $D + c - m$. The full relation quotient embeds in that subspace, and the small-prime equations can only decrease dimension. Therefore

$$\tau \leq D + c - m = (D - d) + (c - n) = D^\# + c^\#.$$

For $m < 2$, the estimate is the direct inclusion $\mathcal{R} \subset W_{>B}$. The inequality $D^\# + 2c^\# \leq 2B$ follows from $D + 2c \leq 2B$. The defect estimate follows from proposition 2.4, applied to each window on its own dyadic slice; (3.1) makes the constants uniform. $\square$

From each component of $\mathcal{C}_{\text{res}}$, choose canonically its smallest large-prime label and then the lexicographically first edge carrying that label. This gives

$$(i_s, j_s, p_s), \quad 1 \leq s \leq c^\#,$$

with distinct offsets in each block, distinct primes $p_s > B$, and $p_s \mid x + i_s, p_s \mid y + j_s$. When $m \geq 2$, these cells are not exact; when $m = 1$, only the at most two components meeting the two occurrences of the unique exact unit can be exceptions. Put

$$P^\# = \prod_{s=1}^{c^\#} p_s,$$

with the empty product equal to one.

The dimension decomposition gives the exact identity

$$2^\rho - 1 = (2^\sigma - 1) + 2^\sigma(2^\tau - 1). \quad (3.6)$$

We first sum the rational term. The residual term will be divided into exhaustive geometric sectors.

3.2 Hosts and rational mass

The elementary prime and divisor estimates used below are

$$\begin{aligned} \pi(B) &\ll B/\log B, & \sum_{p \leq B} p^{-1/2} + B \sum_{p > B} p^{-3/2} &\ll \sqrt{B}/\log B, & (3.7) \\ \sum_{p > B} p^{-2} &\ll (B \log B)^{-1}, & \omega(n) + \log \tau(n) &\ll_K \log M / \log \log M \quad (1 \leq n \leq M^K). & (3.8) \end{aligned}$$

Here $\tau(n)$ in (3.8) is the divisor function, not the residual dimension. The last bound is used only for fixed K . Partial summation of the prime number theorem proves the prime estimates; the divisor estimate is standard [34, 26]. In particular,

$$\prod_{p \leq B} (1 + cp^{-1/2}) = M^{o(1)}, \quad 2^{\pi(B)} = M^{o(1)} \quad (c \text{ fixed}), \quad (3.9)$$

while a squarefree B -smooth integer has size at most

$$\prod_{p \leq B} p = \exp(O(B)) = M^{O_{\beta+}(1)}. \quad (3.10)$$

Let $H_{2,\delta}(M, L)$ count separated ordered pairs in U^2 with $\rho_L(x, y) > 0$. We also need $H_{2,\delta}^{\text{val}}(M, B)$, the number of such pairs for which some nonempty subset of $V_x \sqcup V_y$ has square product, with no parity requirement in either window.

Proposition 3.7 (Host bound). *Uniformly in the fixed band,*

$$H_{2,\delta}(M, L) \leq H_{2,\delta}^{\text{val}}(M, B) \leq M^{3/2+o(1)}.$$

Proof. For each pair counted by $H_{2,\delta}^{\text{val}}$, choose a nonempty square-product subset and its first selected occurrence, ordering first the x -block, then the y -block, and then the offsets increasingly. Suppose first that this occurrence is $n = x + i$; the other orientation costs a factor two. Put

$K = K_B(n)$. For every $p \mid K$, the parity of the selected square product forces an occurrence of odd p -valuation in the other block. It is unique there because $p > B$ and the block diameter is $B - 1$. Forgetting compatibility between primes, there are at most $B^{\omega(K)}$ assignments of the primes of K to offsets of the other block.

A fixed assignment determines one residue class for y modulo every $p \mid K$, hence one class modulo K . The interval $U_{M,\delta}$ has length less than M , so it contains at most $M/K + 1$ representatives. By (3.2), $K \leq n \leq 2M$, and therefore

$$M/K + 1 \leq 3M/K,$$

including $K = 1$. The orientation and selected offset cost at most $2B$, and hence

$$H_{2,\delta}^{\text{val}}(M, B) \ll BM \sum_{n \leq 2M} \frac{B^{\omega(K_B(n))}}{K_B(n)}. \quad (3.11)$$

Write uniquely $n = a^2 ur$, where u is squarefree and B -smooth and $r = K_B(n)$ is squarefree with all prime factors $> B$. Summing over a first gives

$$\begin{aligned} \sum_{n \leq 2M} \frac{B^{\omega(K_B(n))}}{K_B(n)} &\leq \sqrt{2M} \sum_u u^{-1/2} \sum_r B^{\omega(r)} r^{-3/2} \\ &\leq \sqrt{2M} \prod_{p \leq B} (1 + p^{-1/2}) \prod_{p > B} (1 + Bp^{-3/2}). \end{aligned}$$

For this host estimate, elementary integer sums suffice:

$$\sum_{p \leq B} p^{-1/2} \leq \sum_{n=1}^B n^{-1/2} \leq 2\sqrt{B}, \quad B \sum_{p > B} p^{-3/2} \leq B \int_B^\infty t^{-3/2} dt = 2\sqrt{B}.$$

Since $\log(1+t) \leq t$, the weighted kernel sum is at most $\sqrt{2M} \exp(4\sqrt{B})$. Under $B \leq \beta_+ \log M$, $B \exp(4\sqrt{B}) = M^{o(1)}$, uniformly in the length. Substitution in (3.11) proves the claim. The sharper prime-sum bounds are not needed at this step. $\square$

Proposition 3.8 (Rational mass). *The contributions with canonical height $q = \max(a, b)$ satisfy*

$$\sum_{q \geq 3} (2^\sigma - 1) \leq MQ_B^{1/3} M^{o(1)}, \quad \sum_{q=2} (2^\sigma - 1) \leq MQ_B^{1/2} M^{o(1)}. \quad (3.12)$$

The geometry-only sum, without translation parameters, satisfies

$$\sum_{\substack{(a,b)=1, \max(a,b) \geq 2, \\ m(a,b,h) \geq 2}} 2^{m(a,b,h)-1} \leq Q_B^{1/2} M^{o(1)}. \quad (3.13)$$

Proof. At height q there are $O(q)$ reduced pairs (a, b) and $O(qB)$ possible nonempty values of h . A unit (i_0, j_0) gives $x = bt - i_0$, $y = at - j_0$, with $O(M/q)$ choices of t . Units are spaced by (b, a) , so $m - 1 \leq B/q$, and $m \geq 2$ implies $q \leq L$. Summing $O(MqB^{2B/q})$ for $q \geq 3$ gives the first bound; $q = 2$ gives the second. At $q = 1$, separation excludes a unit. Omitting the $O(M/q)$ translation factor and summing the same geometries proves (3.13). $\square$

Remark 3.9 (Why the dyadic interval helps, and what it does not remove). The starts $(x, 2x)$ cannot both belong to $[N, 2N]$. This removes the volumetric $2 : 1$ family, not all ratio-two occurrences in two windows: $(N - 1, 2N - 2)$ can still occur when the starts are N and $2N - 1$.

Our master estimate keeps the volumetric branch and therefore also applies to the finite prefix. In a dyadic restriction, a more precise channel count improves this particular term, but that improvement is not needed for (3.4).

3.3 The partition and the shallow sectors

For a nonaligned pair let $\tilde{k}$ be the rank of the small-prime and two block-parity equations after solving all large-prime equations. Then

$$\tau = D^\# + c^\# - \tilde{k}. \quad (3.14)$$

Set $T_B = \lfloor (B-2)/3 \rfloor$. After removing $2^\sigma - 1$ in (3.6), apply the following tests successively:

Sector	Defining test, after the preceding tests have failed
1	$P^\# \leq M$.
2	$\sigma > 0$ and the canonical channel has $q \leq \sqrt{\log B}$.
3	$c^\# \leq B/6$.
4	The pair is aligned.
5	$c^\# \leq 2B/3$.
6	$D^\# \geq 3$.
7	Write $c^\# = B - s$; then $s + \tilde{k} > T_B$.
8	$s + \tilde{k} \leq T_B$.

The tests are complementary at each step. In sectors 5–8 the pair is nonaligned, $\sigma = 0$, and $D^\# = D$, $c^\# = c$. In sectors 7–8 one also has $c^\# > 2B/3$ and $D^\# \leq 2$. Thus this is a disjoint partition of all residual mass, including pairs with very different heights.

Proposition 3.10 (Direct bounds in sectors 1–3). *Up to a common factor $M^{o(1)}$, the residual masses of sectors 1, 2 and 3 are bounded, respectively, by*

$$M^{3/2} + MQ_B^{1/2}, \quad M^{3/2} + MQ_B^{1/2} + MQ_B^{1/3}, \quad M^{3/2}Q_B^{1/6}.$$

Proof. In sector 1 the canonical certificate contains exactly $c^\#$ distinct primes, all greater than B . Hence

$$(B+1)^{c^\#} \leq P^\# \leq M, \quad c^\# \leq \frac{\log M}{\log(B+1)}, \quad 2^{c^\#} = M^{o(1)}. \quad (3.15)$$

The last estimate is uniform in the fixed positive logarithmic band. Together with $D^\# = O(B/\log B)$ and $\tau \leq D^\# + c^\#$, it gives $2^\tau = M^{o(1)}$ pointwise. When $\sigma = 0$, only relational hosts contribute, and their number is $M^{3/2+o(1)}$ by proposition 3.7. When $\sigma > 0$, use

$$2^\sigma(2^\tau - 1) \leq 2(2^\sigma - 1)M^{o(1)}$$

and sum the rational mass in proposition 3.8. The two branches give $M^{o(1)}(M^{3/2} + MQ_B^{1/2})$, without a critical-scale substitution or any restriction on the ratio between the starts.

In sector 2, write the canonical channel as $h = by - ax$. If a prime $p > B$ joins a nonexact cell (i, j) , it divides $a(x+i) - b(y+j) = ai - bj - h \neq 0$. The existence of an exact unit bounds this integer by $O(qB)$. Thus every residual certificate prime is at most a fixed multiple of qB . Consequently $c^\# \leq \pi(CqB) = o(B)$ uniformly for $q \leq \sqrt{\log B}$. Also $D^\# = o(B)$, so $\tau = o(B)$. Summing the systematic weights by (3.12) bounds this sector. The displayed $M^{3/2}$ term is harmless and also covers a partition convention that sends a zero-dimensional rational code here.

In sector 3, $\sigma = 0$ or $q > \sqrt{\log B}$, hence $\sigma = o(B)$. The quotient estimate gives $2^\sigma(2^\tau - 1) \leq Q_B^{1/6}M^{o(1)}$. Only relational hosts contribute. Multiply by their $M^{3/2+o(1)}$ count. $\square$

The untreated pairs now have a large certificate product and more than $B/6$ residual components. The next step either rules out their alignment or extracts a bounded component whose square class supplies a rigid Diophantine equation.

3.4 Diophantine counting and deep cores

A bounded component forces a product of a fixed number of shifted values to lie in one square class. We need to count solutions only in a polynomial height range. Two of the factors already suffice: after localizing their square classes, they satisfy a Pell equation. This bounded-height argument keeps the uniformity we need without a height-free integral-point theorem.

Lemma 3.11 (Uniform Pell count). *Fix $K_0 > 0$. Let A, C be positive squarefree integers with $A/C \notin \mathbb{Q}^{\times 2}$, and let $e \neq 0$. If $A, C, |e| \leq M^{K_0}$, then the number of integer solutions of*

$$Az^2 - Cw^2 = e, \quad |z|, |w| \leq M^{K_0},$$

is

$$\exp\left(O_{K_0}\left(\frac{\log M}{\log \log M}\right)\right) = M^{o(1)}.$$

Proof. The detailed reduction and uniformity checks are given in [Lemma A.2](#). $\square$

Lemma 3.12 (Split products in a polynomial range). *Fix $d \geq 2$ and $K > 0$. Let $h_1, \dots, h_d$ be distinct integers and e a positive squarefree integer with $|h_i|, e \leq M^K$. The number of integer pairs (X, Y) satisfying*

$$|X| \leq M^K, \quad X + h_i > 0 \ (1 \leq i \leq d), \quad \prod_{i=1}^d (X + h_i) = eY^2$$

is at most $\exp(O_{d,K}(\log M / \log \log M))$.

Proof. Put $\Delta = \prod_{i < j} (h_i - h_j) \neq 0$ and $S = \{p : p \mid e\Delta\}$. If $p \notin S$, it divides at most one factor $X + h_i$ and has even valuation in the product. Hence the squarefree part of every factor is supported on S . In particular

$$X + h_1 = su^2, \quad X + h_2 = tv^2, \quad su^2 - tv^2 = h_1 - h_2 \neq 0,$$

where s, t are positive squarefree integers supported on S . There are at most $4^{|S|}$ choices of (s, t) . We need retain only $s, t \leq 2M^K$, and $u, v \leq (2M^K)^{1/2}$. For $s \neq t$, their quotient is not a rational square, so Lemma 3.11 bounds the solutions uniformly. For $s = t$, the equation $s(u - v)(u + v) = h_1 - h_2$ is bounded by the divisor estimate. Each pair (u, v) determines X , and then at most two values of Y . Finally $|e\Delta| \leq M^{O_{d,K}(1)}$, so $|S| = O_{d,K}(\log M / \log \log M)$; the divisor and Pell bounds have the same exponential scale. Multiplying the bounds proves the claim. $\square$

This argument counts bounded-height solutions, not all integral points. The distinction is precisely what makes a single Pell equation sufficient. The square-class and height bookkeeping is recorded in [section A.3](#).

Lemma 3.13 (Normalization of a component). *Let I, J be two nonempty sets of offsets, with $|I| + |J| \leq K_0$, and let d be a positive squarefree B -smooth integer. If*

$$P_I(x)Q_J(y) = dz^2, \quad P_I(X) = \prod_{i \in I} (X + i), \quad Q_J(Y) = \prod_{j \in J} (Y + j), \quad (3.16)$$

then, after fixing x, I, J, d , there is a positive squarefree integer $e \leq M^{O_{K_0}(1)}$ such that

$$Q_J(y) = ev^2. \quad (3.17)$$

The integer e is the squarefree part of $dP_I(x)$.

Proof. In the group $\mathbb{Q}^\times/\mathbb{Q}^{\times 2}$, the equality (3.16) gives

$$[Q_J(y)] = [dP_I(x)].$$

The positive squarefree representative of the right-hand class is e ; the squarefree part of $Q_J(y)$ is therefore e . Since $|I| \leq K_0$, we have $dP_I(x) \leq M^{O_{K_0}(1)}$. $\square$

Lemma 3.14 (One-sided count of a component). *Fix K_0 . For x, I, J, d fixed as in Lemma 3.13, the number of $y \in U_{M,\delta}$ satisfying (3.16) is*

$$\begin{cases} O(M^{1/2}), & |J| = 1, \\ M^{O_{K_0}(1)}, & |J| \geq 2. \end{cases}$$

The same assertion holds after exchanging the two blocks.

Proof. Lemma 3.13 gives $Q_J(y) = ev^2$. For $|J| = 1$, the equation $y + j = ev^2$ leaves at most $1 + \sqrt{3M/e} = O(M^{1/2})$ values. For $|J| \geq 2$, use Lemma 3.12. Its degree is at most K_0 , all shifted factors are positive, $y \leq M$, and $e \leq M^{O_{K_0}(1)}$. Thus a single fixed polynomial-height exponent covers every coefficient and variable. The symmetric statement follows by interchanging the blocks. $\square$

Lemma 3.15 (Two singleton components). *Fix two offsets i, j . The number of triples (x, y, d) , with $x, y \in U_{M,\delta}$, d positive, squarefree and B -smooth, satisfying*

$$(x + i)(y + j) = dz^2$$

is

$$M \exp\left(O\left(\frac{\sqrt{B}}{\log B}\right)\right). \quad (3.18)$$

Proof. Set $X = x + i$, $Y = y + j$. A prime-by-prime analysis of $XY = dz^2$ gives a unique parametrization

$$X = ecu^2, \quad Y = (d/e)cv^2,$$

where $e \mid d$, where c is squarefree and $(c, d) = 1$. For fixed e , the conditions $1 \leq X, Y \leq 3M$ force

$$c \leq C_e := \frac{3M}{\max(e, d/e)}.$$

Dropping the “squarefree” and “coprime to d ” conditions, which only increases the count, the number of triples (u, v, c) is at most

$$\begin{aligned} \sum_{c \leq C_e} \left(1 + \sqrt{\frac{3M}{ec}}\right) \left(1 + \sqrt{\frac{3Me}{dc}}\right) &\ll C_e + \sqrt{\frac{M}{e}}\sqrt{C_e} + \sqrt{\frac{Me}{d}}\sqrt{C_e} + \frac{M}{\sqrt{d}} \sum_{c \leq C_e} \frac{1}{c} \\ &\ll \frac{M \log M}{\sqrt{d}}. \end{aligned}$$

Here we used $\max(e, d/e) \geq \sqrt{d}$. After summing over $e \mid d$, the number of solutions for fixed d is

$$\ll \frac{M\tau(d) \log M}{\sqrt{d}}.$$

Finally

$$\sum_{\substack{d \text{ squarefree} \\ p \mid d \Rightarrow p \leq B}} \frac{\tau(d)}{\sqrt{d}} = \prod_{p \leq B} \left(1 + \frac{2}{\sqrt{p}}\right) = \exp\left(O\left(\frac{\sqrt{B}}{\log B}\right)\right).$$

The factor $\log M$ is absorbed into the exponential $M^{o(1)}$. $\square$

Lemma 3.16 (Hosts of a bounded component). *Fix K_0 . The number of separated pairs (x, y) whose non-aligned residual graph contains a nontrivial unpinned component meeting both blocks and of size at most K_0 is*

$$M^{1+o_{K_0}(1)}. \quad (3.19)$$

If this component has exactly two vertices, one in each block, we have the sharper bound

$$M \exp\left(O\left(\frac{\sqrt{B}}{\log B}\right)\right).$$

Proof. Let I, J be the offset sets of the component. Lemma 3.5 provides, with a unique coefficient d , equation (3.16). There are only $B^{O_{K_0}(1)} = M^{o(1)}$ choices of the two offset sets. We separate three branches.

Two degrees at least two. If $|I|, |J| \geq 2$, fix x , the offsets and d . Lemma 3.14 leaves $M^{o_{K_0}(1)}$ values of y . Summing over the M values of x and the $2^{\pi(B)} = M^{o(1)}$ coefficients d gives $M^{1+o_{K_0}(1)}$ hosts.

A single degree equal to one. If $|I| = 1 < |J|$, the same argument fixes x and uses the mobile side of degree $|J| \geq 2$. If $|J| = 1 < |I|$, we exchange the two blocks. We again obtain $M^{1+o_{K_0}(1)}$ after summing over d .

Two singletons. If $|I| = |J| = 1$, Lemma 3.15 applies directly. It already counts the triples (x, y, d) and, in particular, performs the simultaneous summation

$$\sum_{d \text{ } B\text{-smooth, squarefree}} \frac{\tau(d)}{\sqrt{d}}.$$

No factor $2^{\pi(B)}$ must be reintroduced in this branch. After the union over the B^2 offset pairs, absorbed into the exponential, the sharp bound remains

$$M \exp\left(O\left(\frac{\sqrt{B}}{\log B}\right)\right).$$

The three branches give both assertions. $\square$

Proposition 3.17 (No aligned residual core of linear density). *For every fixed $\alpha > 0$, no aligned pair in $U_{M,\delta}^2$ satisfies $c^\# \geq \alpha B$ once M is sufficiently large in terms of the fixed parameters and α .*

Proof. Let m_{ex} be the number of exact units of the canonical channel. If $m_{\text{ex}} \geq 2$, the nondefective exact components have already been removed in $\mathcal{C}_{\text{res}}$. If $m_{\text{ex}} = 1$, make no height or valuation assumption and remove directly the at most two graph components containing its two occurrences. If $m_{\text{ex}} = 0$, remove nothing. In every case, from $c^\# \geq \alpha B$ there remain at least $\alpha B/2$ nontrivial components containing no exact unit. Their total size is at most $2B$, so at least $m \geq \alpha B/4$ have size at most $K_\alpha = \lceil 8/\alpha \rceil$.

For each such component C_t form the column

$$\Phi(C_t) = \left((v_p(\prod_{n \in C_t} n) \bmod 2)_{p \leq B}, |C_t \cap V_x| \bmod 2, |C_t \cap V_y| \bmod 2 \right) \in \mathbb{F}_2^{\pi(B)+2}.$$

The kernel of the matrix of these m columns has codimension at most $r = \pi(B) + 2$. Choose

$$t = \left\lceil C_\alpha \frac{B}{\log B \log \log B} \right\rceil$$

with C_α large. If no nonzero kernel word had weight at most $2t$, the Hamming bound would give $\sum_{j \leq t} \binom{m}{j} \leq 2^r$. Since $m \gg_\alpha B$ and $t = o(m)$,

$$\log \binom{m}{t} \geq t \log(m/t) - O(t) = (C_\alpha + o_\alpha(1)) \frac{B}{\log B},$$

which exceeds $r \log 2$ when C_α is fixed sufficiently large. Thus a nonzero word uses

$$O_\alpha \left(\frac{B}{\log B \log \log B} \right)$$

components.

Let I, J be the unions of their offsets. The last two coordinates of Φ make $|I|$ and $|J|$ even; the remaining coordinates, together with the large-prime equations internal to each component, show that

$$\prod_{i \in I} (x + i) \prod_{j \in J} (y + j)$$

is a square. Put $d_0 = |I| + |J|$; then

$$d_0 \ll_\alpha \frac{B}{\log B \log \log B}.$$

For the canonical channel (a, b, h) define

$$G(T) = \prod_{i \in I} (T + ai) \prod_{j \in J} (T + h + bj).$$

There are no root collisions: one would be an exact unit, and all components containing such a unit were removed. At $T = ax$,

$$G(ax) = a^{|I|} b^{|J|} \prod_{i \in I} (x + i) \prod_{j \in J} (y + j)$$

is an integer square. All roots have modulus $O(B^{A+1})$, while $ax \geq x \geq M^\delta$; in particular $ax \geq 2R$ for the root radius $R = O(B^{A+1})$ once M is large. Lemma 2.3 gives

$$ax \leq (Cd_0 B^{A+1})^{2d_0}.$$

The logarithm of the right-hand side is $O_\alpha(B/\log \log B) = o(B)$, whereas

$$\log(ax) \geq \delta \log M \geq (\delta/\beta_+)B.$$

This contradiction closes the aligned deep sector directly at scale M . $\square$

We now work in the nonaligned complement. A component of size at most $2/\alpha$ exists whenever $c^\# \geq \alpha B$, since the total number of occurrences is $2B$. The preceding component count therefore gives $M^{1+o(1)}$ hosts in every fixed positive-density deep sector. For sector 5, $\tau \leq D^\# + c^\# \leq 2B/3 + o(B)$, so its mass is $MQ_B^{2/3}M^{o(1)}$.

Lemma 3.18 (Two defects in one block). *The number of starts $x \in U_{M,\delta}$ whose window contains two distinct B -defective occurrences is $M^{o(1)}$.*

Proof. Write $x + i_1 = d_1u^2$ and $x + i_2 = d_2v^2$, with d_1, d_2 squarefree and B -smooth. If $d_1 \neq d_2$, then d_1/d_2 is not a square and

$$d_1u^2 - d_2v^2 = i_1 - i_2 \neq 0$$

is covered by Lemma 3.11; all coefficients and variables are in fixed polynomial ranges by (3.10) and (3.2). If $d_1 = d_2 = d$, then $d \mid i_1 - i_2$ and

$$(u - v)(u + v) = (i_1 - i_2)/d,$$

which has $M^{o(1)}$ solutions by the divisor bound. There are $4^{\pi(B)}B^2 = M^{o(1)}$ choices of coefficients and offsets. $\square$

Proposition 3.19 (Sector 6). *For nonaligned deep-core pairs with $c^\# > 2B/3$ and $D^\# \geq 3$,*

$$R_{\text{res}} \leq M^{1/2}Q_B M^{o(1)}.$$

Proof. One block contains at least two defects, so its start has $M^{o(1)}$ choices by Lemma 3.18. Since $c^\# > 2B/3$, a residual component has bounded size. Fix its offsets and square class at a cost $M^{o(1)}$. The start in the other block is then counted by Lemma 3.14: at most $M^{1/2}$ choices when its component degree is one, and $M^{o(1)}$ otherwise. Hence the number of hosts is $M^{1/2+o(1)}$.

Finally $D^\# + 2c^\# \leq 2B$ and $D^\# = O(B/\log B)$ imply $D^\# + c^\# \leq B + o(B)$. Therefore $2\tau \leq Q_B M^{o(1)}$, and multiplication proves the result. $\square$

For sector 7, integrality gives $s + \tilde{k} \geq T_B + 1$. Thus (3.14) and $D^\# \leq 2$ imply $\tau \leq B + 1 - T_B = 2B/3 + O(1)$. The same deep-host count bounds its mass by $MQ_B^{2/3}M^{o(1)}$.

3.5 Terminal kernels: count clustering, not just candidates

It remains to consider sector 8. Write $j = s + \tilde{k} \leq T_B$. If a component has size $2 + e_t$, then $\sum_t e_t \leq 2B - 2c^\# = 2s$. At most $2s$ components have size greater than two, so at least $B - 3s \geq B - 3j \geq 2$ are isolated size-two components. For such a component $\{x + i_t, y + j_t\}$ put $R_t = K_B(x + i_t)$.

Lemma 3.20 (Terminal kernels and determinants). *For every isolated size-two component,*

$$R_t = K_B(x + i_t) = K_B(y + j_t) > 1.$$

Kernels belonging to distinct components are coprime. For $s \neq t$,

$$R_s R_t \mid \Delta_{s,t} := (x + i_s)(y + j_t) - (x + i_t)(y + j_s),$$

and

$$0 < |\Delta_{s,t}| \leq 4MB$$

for large M .

Proof. An isolated unpinned component contains exactly the two occurrences carrying every one of its $p > B$ labels, so the full kernels agree and are nontrivial. A prime cannot label two components, because each block contains at most one odd occurrence of that prime; hence the kernels are coprime. Divisibility of the determinant follows term by term.

If the determinant vanished, the two occurrence ratios would be equal. Writing their common reduced ratio as b/a gives $x + i_s = bu_s$, $y + j_s = au_s$ and similarly with u_t . Subtraction yields

$$i_s - i_t = b(u_s - u_t), \quad j_s - j_t = a(u_s - u_t).$$

Distinct components have distinct offsets, so $u_s \neq u_t$, and $a, b \leq L$. The two cells would therefore be two units of an aligned exact channel, contradicting nonalignment. Finally

$$|\Delta_{s,t}| \leq |x + i_s||j_t - j_s| + |y + j_s||i_s - i_t| \leq 4MB$$

by (3.2). □

Slice terminal pairs by $X \leq \max(x, y) < 2X$, where $M^\delta/2 \leq X \leq M$. For counting, call the larger coordinate z and the smaller one its partner; keep each pair's original canonical data. The two possible orientations cost only a factor two, and no symmetry of the canonical selection is assumed. On this slice $R_s R_t \ll XB$. Set $T_X = (CXB)^{1/2}$ with an absolute constant large enough for this inequality. At most one isolated kernel exceeds T_X .

Lemma 3.21 (Partners and a one-dimensional container). *On a terminal slice, a fixed value of either start has $X^{o(1)}$ possible partners. The number of possible values of either coordinate, and hence of the larger start, is at most $X^{3/4+o(1)}$. All constants are uniform in the slice.*

Proof. Fix either start and write it as x , with partner y , for this count only; the original canonical data of the pair are unchanged. Choose two isolated components at a cost $O(B^4) = X^{o(1)}$. Their distinct coprime kernels $R, S > 1$ are fixed by x . The other occurrences have the unique forms $y + j = Ruz^2$, $y + v = Su'w^2$ with independent squarefree B -smooth coefficients u, u' . Thus

$$Ruz^2 - Su'w^2 = j - v \neq 0.$$

The coefficients Ru, Su' are squarefree, of size $O(X)$, and their quotient is nonsquare because the large-prime supports of R, S are disjoint. The Pell bound gives $X^{o(1)}$ partners per pair of smooth coefficients, of which there are $4^{\pi(B)} = X^{o(1)}$. This argument needs only $y \ll X$, not $y \asymp X$.

At least one isolated kernel is at most T_X . Integers with such a kernel have the unique representation $n = ua^2r$, where u is squarefree and B -smooth and $r = K_B(n) \leq T_X$. Therefore

$$\#\{n \leq 3X : K_B(n) \leq T_X\} \leq \sqrt{3X} \prod_{p \leq B} (1 + p^{-1/2}) \sum_{r \leq T_X} r^{-1/2} \ll X^{3/4+o(1)}. \quad (3.20)$$

Each such integer lies in at most $B = X^{o(1)}$ windows, whether the coordinate is the first or the second start. Their union also bounds the larger starts. Since $B/\log X$ stays in a fixed compact interval, all constants are uniform. $\square$

Lemma 3.22 (Shifted small-kernel energy). *For $T \ll (XB)^{1/2}$ put $A_T(x) = \#\{i \in I_L : K_B(x+i) \leq T\}$. Uniformly on every slice,*

$$\sum_{X \leq x < 2X} \binom{A_T(x)}{2} \ll_{\varepsilon} X^{2/3+\varepsilon} B^2. \quad (3.21)$$

Proof. Let $W_B = \prod_{p \leq B} (1 + p^{-1/2}) = X^{o(1)}$. Kernels $r \leq R_0 = X^{1/3}$ occur at at most

$$\sqrt{3X} W_B \sum_{r \leq R_0} r^{-1/2} = X^{2/3+o(1)}$$

integers in $[X/2, 3X]$. Each can anchor at most $O(B^2)$ pairs of occurrences and windows, so these integers give the required bound.

For the remaining pairs fix $1 \leq h \leq B$ and dyadic ranges $r = K_B(n) \asymp R$, $s = K_B(n+h) \asymp S$, with $R, S \geq X^{1/3}$ and $R, S \ll (XB)^{1/2}$. Set $q = n/r$, $q' = (n+h)/s$. Both are a square times a squarefree B -smooth integer. There are at most $X^{o(1)} \sqrt{X/R}$ and $X^{o(1)} \sqrt{X/S}$ choices, respectively. Moreover $q \asymp X/R$ and $q' \asymp X/S$, because $n \asymp X$. For a fixed pair, the equation

$$qr + h = q's$$

requires $(q, q') \mid h$; if soluble, its integer solutions have steps $q'/(q, q')$ and $q/(q, q')$. Hence at most $1 + O(BRS/X)$ solutions lie in the specified boxes. Their total is

$$X^{o(1)} \left(\frac{X}{\sqrt{RS}} + B\sqrt{RS} \right) \leq X^{2/3+o(1)}.$$

There are $X^{o(1)}$ pairs of dyadic ranges. Sum over h and note that a shifted pair lies in at most B windows. This proves (3.21). $\square$

Proposition 3.23 (Terminal mass). *Sector 8 contributes at most*

$$M^{\varepsilon} \left(M^{2/3} Q_B + M^{3/4} Q_B^{2/3} \right).$$

Proof. On a fixed larger-start slice put $j = s + \tilde{k}$ and $m_j = B - 3j - 1$. The sector test gives $m_j \geq 1$. By equation (3.14), $\tau + j = B + D^{\#}$ with $D^{\#} \leq 2$. There are at least $B - 3j$ isolated kernels, all but at most one below T_X , so each window has at least m_j small kernels.

First collect all pairs with $m_j \geq 2$, without separating the values of j . Their larger start z satisfies $\binom{A_{T_X}(z)}{2} \geq 1$. The energy estimate counts at most $X^{2/3+o(1)} B^2$ such starts, and the uniform partner bound contributes only $X^{o(1)}$. Since $2^{\tau} - 1 \leq 4Q_B$, the total mass is at most $X^{2/3+o(1)} Q_B$; the factor B^2 is subpolynomial.

In the complementary population, $m_j \leq 1$ implies $3j \geq B - 2$. Consequently $3\tau \leq 2B + 8$, and $2^{\tau} - 1 \leq 8Q_B^{2/3}$. At least one small kernel remains, so the one-dimensional container and partner bound give at most $X^{3/4+o(1)}$ pairs. Their mass is at most $X^{3/4+o(1)} Q_B^{2/3}$.

The dyadic slices of the larger coordinate exhaust the terminal pairs. Their logarithmic bands are uniform, and geometric summation proves the claim. The partner estimate uses only its upper height bound, never comparability with X . $\square$

3.6 Assembly and lower bounds

Proof of Theorem 3.1. The completed estimates are as follows; a factor $M^{o(1)}$ is suppressed.

Population	Mechanism	Weighted mass
Rational	Exact-unit enumeration	$MQ_B^{1/3} + MQ_B^{1/2}$
1	Small product and rational mass	$M^{3/2} + MQ_B^{1/2}$
2	Small channel height	$M^{3/2} + MQ_B^{1/2} + MQ_B^{1/3}$
3	Host count and rank	$M^{3/2}Q_B^{1/6}$
4	Runge and a short codeword	0
5	Bounded components	$MQ_B^{2/3}$
6	Two defects and a component	$M^{1/2}Q_B$
7	Exact rank deficit	$MQ_B^{2/3}$
8	Shifted kernel energy and Pell	$M^{2/3}Q_B + M^{3/4}Q_B^{2/3}$

Every entry is bounded by one of the three monomials in (3.3). The partition is exhaustive and (3.6) is exact, so summation proves that profile. Finally

$$M^{3/2}Q_B^{1/6} = (M^{5/3})^{5/6}(M^{2/3}Q_B)^{1/6}, \quad MQ_B^{2/3} = (M^{5/3})^{1/3}(M^{2/3}Q_B)^{2/3},$$

weighted arithmetic–geometric mean proves (3.4). $\square$

The exponent $5/3$ is an upper bound, not an optimality claim. In a dyadic block, the ordered pairs $x = 2t, y = 3t$, $N/2 \leq t < 2N/3$, and their reversals give

$$R_2(N, L) \geq \left(\frac{N}{3} + O(1)\right) \left(2^{\lfloor (B+1)/3 \rfloor - 1} - 1\right). \quad (3.22)$$

On $U_{M,\delta}$, the pairs $x = t, y = 2t$ and their reversals give

$$R_{2,\delta}(M, L) \geq (M - 2M^\delta + O(1)) \left(2^{\lfloor B/2 \rfloor - 1} - 1\right). \quad (3.23)$$

These follow simply by counting the even code of exact units, regardless of any extra residual relations. At criticality the lower exponents are $4/3$ and $3/2$, respectively. The one-dimensional container (3.20) is itself of order $X^{3/4+o(1)}$ when $T \asymp \sqrt{XB}$: choose r prime near T and $n = ra^2 \asymp X$. What improves the terminal estimate is not a smaller container, but the fact that its elements cannot cluster sufficiently often in one short window, as quantified by (3.21).

The weighted relation estimate is now available. A Poisson comparison still requires the conditional marginal probabilities, an exact dependency graph and the local overlap counts. Before constructing that graph, we record two forms of the estimate adapted to prescribed words.

3.7 From relative signs to prescribed values

There is little extra arithmetic in prescribing the sign of a word rather than identifying it with its negative. What changes is the parity restriction on square-product subsets.

For separated windows let $\rho_B^{\text{val}}(x, y)$ be the nullity of the transpose of the matrix with all $2B$ valuation rows $v(n)$, $n \in V_x \sqcup V_y$. Thus its kernel consists of all square-product subsets, without requiring even cardinality in each block. Write $R_{2,\delta}^{\text{val}}(M, B)$ for the sum of $2\rho_B^{\text{val}}(x, y) - 1$ over the same separated pairs as $R_{2,\delta}$, and $R_2^{\text{val}}(N, B)$ for its dyadic analogue.

Proposition 3.24 (Prescribing the global signs). *Uniformly under the hypotheses of Theorem 3.1,*

$$R_{2,\delta}^{\text{val}}(M, B) \leq 4R_{2,\delta}(M, B-1) + 3H_{2,\delta}^{\text{val}}(M, B), \quad (3.24)$$

$$R_{2,\delta}^{\text{val}}(M, B) \ll_{\varepsilon} M^{\varepsilon} \left(M^{3/2} Q_B^{1/6} + M Q_B^{2/3} + M^{2/3} Q_B \right) \ll_{\varepsilon} M^{\varepsilon} (M^{5/3} + M^{2/3} Q_B). \quad (3.25)$$

The same conclusion holds on a dyadic block and on bounded-ratio subintervals, with the corresponding ambient scale.

Proof. Imposing the two block-parity equations on the full square-relation kernel gives exactly the relative-sign kernel, by the tree boundary bijection. Therefore

$$\rho_{B-1}(x, y) \leq \rho_B^{\text{val}}(x, y) \leq \rho_{B-1}(x, y) + 2.$$

If the full kernel is zero both weights vanish. Otherwise

$$2^{\rho_B^{\text{val}}} - 1 \leq 4(2^{\rho_{B-1}} - 1) + 3.$$

The host term includes every separated ordered pair admitting a nonempty square-product subset of the full vertex occurrences, with no even-cardinality condition in either block. In particular it may count a pair whose relative-sign kernel is zero. Sum the inequality using exactly this unrestricted host estimate from proposition 3.7. Its $M^{3/2+o(1)}$ term is absorbed by the first monomial in (3.3). Restriction proves the other geometries. The extra two affine constraints therefore cost only a fixed factor and a sparse set of new relational pairs, not an exponential factor in the window length. $\square$

3.8 Capping the relation weight

The full relation weight is useful when estimating moments. For the probability that one of many mutually exclusive words occurs, however, its largest values can be wasteful: a joint probability cannot exceed either marginal. We record a version of the arithmetic estimate that preserves this elementary ceiling.

For $T \geq 1$ define

$$R_{2,\delta}^{[T]}(M, L) = \sum_{\substack{x, y \in U_{M,\delta} \\ |x-y| > L}} \min\{T, 2^{\rho_L(x,y)} - 1\}.$$

The notation $R_{2,\delta}^{\text{val},[T]}(M, B)$ has the same meaning with the full-value nullity ρ_B^{val} ; omission of δ denotes the corresponding dyadic sum.

Proposition 3.25 (Capped two-window profile). *Under the fixed-band hypotheses of Theorem 3.1, uniformly also for $T \geq 1$,*

$$R_{2,\delta}^{[T]}(M, L) \ll_{\varepsilon} M^{\varepsilon} \left(M^{3/2} Q_B^{1/6} + M Q_B^{2/3} + M^{2/3} \min\{T, Q_B\} \right), \quad (3.26)$$

$$R_{2,\delta}^{\text{val},[T]}(M, B) \ll_{\varepsilon} M^{\varepsilon} \left(M^{3/2} Q_B^{1/6} + M Q_B^{2/3} + M^{2/3} \min\{T, Q_B\} \right). \quad (3.27)$$

The same bounds hold on dyadic and bounded-ratio subintervals. The implicit constants are independent of T .

Proof. Write $2^{\rho} - 1 = (2^{\sigma} - 1) + 2^{\sigma}(2^{\tau} - 1)$ and use $\min(T, a + b) \leq a + \min(T, b)$ for $a, b \geq 0$. The rational contribution and sectors 1–5 and 7 are left uncapped; the assembly table bounds their sum by $M^{o(1)}(M^{3/2} Q_B^{1/6} + M Q_B^{2/3})$. In sector 6, the proof of its host bound gives $M^{1/2+o(1)}$

pairs, and the pointwise weight is at most $Q_B M^{o(1)}$. Here the factor is chosen independently of T . The inequality $\min(T, EQ_B) \leq E \min(T, Q_B)$ for $E \geq 1$ therefore bounds this contribution by $M^{1/2+o(1)} \min(T, Q_B)$.

For sector 8, use the two populations in the proof of proposition 3.23. On a larger-start slice, the first population has at most $X^{2/3+o(1)}$ pairs and pointwise weight at most $4Q_B$. All geometric and partner bounds were fixed before T , so

$$\min\{T, 2^\tau - 1\} \leq 4 \min\{T, Q_B\}$$

gives capped mass $X^{2/3+o(1)} \min\{T, Q_B\}$ with a common subpolynomial factor independent of T . Leave the complementary population uncapped: its mass $X^{3/4+o(1)} Q_B^{2/3}$ is absorbed by $X^{1+o(1)} Q_B^{2/3}$. Geometric summation proves (3.26).

Finally, the two parity constraints give the pointwise inequality

$$\min\{T, 2^{\rho_B^{\text{val}}} - 1\} \leq 4 \min\{T, 2^{\rho_B - 1} - 1\} + 3 \mathbf{1}_{\{\rho_B^{\text{val}} > 0\}}.$$

The last indicator has sum $M^{3/2+o(1)}$ by the full-value host bound. This proves (3.27); restriction proves the other geometries. $\square$

The cap changes only the dense terminal contribution. Its benefit will come from choosing T according to a probability, rather than from improving the uncapped exponent $5/3$.

4 From arithmetic relations to Poisson laws

The arithmetic estimate is now available independently of the probability argument. The remaining difficulty is not a lack of pair information, but the absence of a sparse unconditional dependency graph. Conditioning on the small prime signs supplies exactly such a graph.

4.1 An explicit transfer inequality

Let $\mathcal{F}_Y = \sigma(X_p : p \leq Y)$, with $Y > 2B$. Let D_Y be the starts in I_N for which some vertex of V_x is Y -defective, and put $G_Y = I_N \setminus D_Y$. We deliberately use the whole support in this removal rule, so that the same convention works for marks and overlaps. For $x \in G_Y$, the private-pivot lemma gives, for every environment,

$$\mathbb{P}(A_x X = b \mid \mathcal{F}_Y) = 2^{-L} \quad \text{for every } b \in \mathbb{F}_2^L. \quad (4.1)$$

After conditioning, an event at x depends only on the prime coordinates

$$\Pi_x(Y) = \{p > Y : \exists n \in V_x, v_p(n) \text{ odd}\}.$$

Join starts whose coordinate sets intersect. Disjoint collections with no edge between them use disjoint independent coordinates. This is an exact conditional dependency graph, not an approximate decorrelation assertion.

Write $E_Y(A)$ for its ordered edge count restricted to a mask A , and set

$$\mathcal{M}_B(A) = \sum_{x \in A} (2^{m_B(x)} - 1), \quad D_Y(A) = A \cap D_Y, \quad \mathcal{M}_B = \mathcal{M}_B(I_N).$$

All defect counts use the complete support, including $x - 1$. By Corollary 2.5, the actual probability mass removed from A is at most $p(\mathcal{M}_B(A) + |D_Y(A)|)$, where $p = 2^{-L}$. Deleting

the corresponding Poisson target coordinates costs at most $p|D_Y(A)|$. The edge count obeys

$$E_Y(I_N) \ll B^2 \left(\frac{N^2}{Y \log Y} + N \log \log N + \frac{N}{\log N} \right). \quad (4.2)$$

Indeed a prime $p > Y$ can occur in at most $O(B(N/p + 1))$ windows; sum the square of this quantity over $Y < p \leq 3N$ and use partial summation.

We use the dependency-graph form of the Arratia–Goldstein–Gordon theorem [1, 2]. For indicators (I_α) with neighbourhoods B_α containing α , let

$$b_1 = \sum_{\alpha} \sum_{\beta \in B_\alpha} p_\alpha p_\beta, \quad b_2 = \sum_{\alpha} \sum_{\substack{\beta \in B_\alpha \\ \beta \neq \alpha}} \mathbb{E}(I_\alpha I_\beta), \quad \mu = \sum_{\alpha} p_\alpha.$$

An exact dependency graph has no third error term. The scalar Poisson distance is at most $C(1 \wedge \mu^{-1})(b_1 + b_2)$; the joint indicator-to-independent-Poisson distance is at most $C(b_1 + b_2)$. These assertions also apply on every fibre of $\mathcal{F}_Y$.

Theorem 4.1 (Masked arithmetic-to-Poisson transfer). *For every deterministic $A \subset I_N$, put $\mu_A = |A \cap G_Y|p$ and $R_2(A) = \sum_{x,y \in A, |x-y|>L} (2^{\rho(x,y)} - 1)$. Then*

$$\begin{aligned} & \mathbb{E} d_{\text{TV}} \left(\mathcal{L} \left(\sum_{x \in A} J_{x,L} \mid \mathcal{F}_Y \right), \text{Pois}(|A|p) \right) \\ & \ll p(\mathcal{M}_B(A) + 2|D_Y(A)|) + (1 \wedge \mu_A^{-1})p^2(|A| + E_Y(A) + R_2(A)). \end{aligned} \quad (4.3)$$

The multiplier is interpreted as one if $\mu_A = 0$. Replacing it by one gives a valid joint-field bound on the same finite site lattice.

Proof. Delete bad actual sites and their target coordinates. The natural coupling costs at most $p(\mathcal{M}_B(A) + 2|D_Y(A)|)$. On each remaining fibre, the marginals are exactly p , and the graph above is a dependency graph. Thus $b_1 \leq p^2(|A| + E_Y(A))$. Overlapping starts are incompatible. At distance L , the union is a tree with all its vertices possessing private coordinates above Y , since its diameter is $2L < Y$. Its joint probability is therefore p^2 . For separated pairs the affine Fourier identity gives, after averaging,

$$\mathbb{E} b_2 \ll p^2(|A| + E_Y(A) + R_2(A)).$$

Apply the scalar or process theorem on each fibre, average, and add the deletions. The target is deterministic, so mixing also gives the same unconditional bound. $\square$

The denominator uses the mask's own mean. For a sparse mask it cannot be replaced by the ambient intensity Np . At bounded ambient intensity, all nonnegative sums are bounded by their full-block values, so the critical estimate is uniform over all deterministic masks.

4.2 The two prime-cutoff scales

The leading scales are already visible without any special functions. Set $S_N = \sqrt{\log N \log \log N}$ and $Y = e^{aS_N}$. The normalized number of bad starts has leading upper exponent $1/(2a)$, while the normalized edge count has exponent a . Their balance is $a = 1/\sqrt{2}$. For rare conditioning events and growing intensities we retain the smaller second-order terms. The following definitions keep these terms explicit.

For $\nu \geq e$, let $u(\nu) \geq 1$ solve $e^u/u = \nu$, and define

$$D(\nu) = \nu u(\nu) - \text{Ei}(u(\nu)), \quad H_N = \log N.$$

For sufficiently large N , define V_N and $\widehat{V}_N$ by

$$V_N = D(H_N/V_N), \quad \widehat{V}_N = 2D(H_N/\widehat{V}_N), \quad \nu_N = H_N/V_N, \quad \widehat{\nu}_N = H_N/\widehat{V}_N. \quad (4.4)$$

The relevant solutions have $H_N/V_N, H_N/\widehat{V}_N \geq e$. Since $D'(\nu) = u(\nu) > 0$, each equation has exactly one such solution for large N . Put $Y_N^\sharp = \lfloor e^{V_N} \rfloor$ and $\widehat{Y}_N = \lfloor e^{\widehat{V}_N} \rfloor$.

Proposition 4.2 (Uniform cutoff estimates). *Let $w \asymp \sqrt{H_N \log H_N}$, $\nu_w = H_N/w$ and $Y = \lfloor e^w \rfloor$. Uniformly in a fixed logarithmic band,*

$$\frac{|D_Y|}{N} \leq e^{-D(\nu_w)+o(\nu_w)}, \quad \frac{E_Y(I_N)}{N^2} \leq e^{-w+o(\nu_w)} + N^{-1+o(1)}. \quad (4.5)$$

The maximum degree Δ_Y of the graph on all starts, including bad ones, satisfies the same second bound after normalization by N . Moreover,

$$V_N^2 = \frac{H_N}{2} (\log H_N + \log \log H_N - \log 2 - 2 + o(1)), \quad (4.6)$$

$$\widehat{V}_N^2 = H_N (\log H_N + \log \log H_N - \log 4 - 2 + o(1)). \quad (4.7)$$

In particular $\widehat{V}_N \sim \sqrt{2}V_N$ and $\log \log N = o(\nu_N)$.

Proof. If $A(X, Y)$ counts Y -defective integers up to X , their unique square-smooth factorization gives, for $0 < \zeta < 1/2$,

$$A(X, Y)/X \leq X^{-\zeta} \prod_{p \leq Y} (1 + p^{-1+\zeta}).$$

Choose $\zeta = u(\nu_w)/w$. The prime number theorem and partial summation give $\sum_{p \leq e^w} \log(1 + p^{-1+\zeta}) = \text{Ei}(u(\nu_w)) + o(\nu_w)$. This proves the first bound, because $|D_Y| \leq BA(3N, Y)$ and $\log B = o(\nu_w)$. The edge bound follows from (4.2). Each window uses $O(BH_N/w)$ primes above Y , each meeting $O(B(N/Y + 1))$ other windows, which proves the maximum-degree bound. The expansions follow by substituting the asymptotic expansion of $\text{Ei}(u)$ into (4.4). [Proposition B.1](#) gives the uniform remainder calculation and both expansions in detail. $\square$

Theorem 4.3 (Hard deletion and soft retention). *Fix $0 < \beta_- < \beta_+ < \infty$. Uniformly when $\beta_- \log N \leq L + 1 \leq \beta_+ \log N$, for every $\varepsilon > 0$,*

$$d_{\text{TV}}(\mathcal{L}(Z_{N,L}), \text{Pois}(\lambda_N)) \ll \min \left\{ 1, \lambda_N \left(e^{-V_N+o(\nu_N)} + N^{-1/3+\varepsilon} \right) \right\}, \quad (4.8)$$

$$d_{\text{TV}}(\mathcal{L}(Z_{N,L}), \text{Pois}(\lambda_N)) \ll \min \left\{ 1, e^{-(\widehat{V}_N - \log^+ \lambda_N)/2+o(\widehat{\nu}_N)} + N^{-1/3+\varepsilon} \right\}. \quad (4.9)$$

The first estimate also bounds the mean conditional distance given $\mathcal{F}_{Y_N^\sharp}$, and the second given $\mathcal{F}_{\widehat{Y}_N}$. Neither saddle is asserted to be optimal for the distribution itself.

Proof. At the hard cutoff, $|D_Y| = o(N)$ and $\mu_{I_N} \asymp \lambda_N$. The master estimate and $Q_B = 2N/\lambda_N$ give

$$R_2(N, L)/N^2 \ll N^{-1/3+\varepsilon} (1 + \lambda_N^{-1}).$$

Its contribution to (4.3) is $O(\lambda_N N^{-1/3+\varepsilon})$. The defect mass contributes $\lambda_N N^{-1/2+o(1)}$ and the cutoff terms contribute $\lambda_N e^{-V_N+o(\nu_N)}$. This proves (4.8).

For the second estimate retain the exceptional indicators in the scalar sum instead of coupling all of them to zero. The scalar Stein equation has factors $\|f\|_\infty \ll 1 \wedge \lambda^{-1/2}$ and $\|\Delta f\|_\infty \leq 1 \wedge \lambda^{-1}$ [20, 29]. Telescope only the good-start terms, using their exact conditional marginal. Bound the exceptional part by its total first moment times $\|f\|_\infty$. Exceptional neighbours of a good start stay in the neighbour sums; their mean is controlled after averaging, not pointwise. Set $\ell = \log^+ \lambda$ and $K = \max(1, \lambda) = e^\ell$. The same two Stein factors, at the same cutoff, give for every $\lambda > 0$ the upper bound

$$e^{\ell/2 - D(H_N/w) + o(H_N/w)} + e^{\ell - w + o(H_N/w)} + KN^{-1/3 + \varepsilon/2}. \quad (4.10)$$

Here a smaller arithmetic exponent absorbs the subpolynomial terms. The detailed calculation is in [section B.3](#). At $w = \hat{V}_N$ choose a common nonnegative remainder $\eta_N \hat{v}_N$, $\eta_N \rightarrow 0$, in the two exponentials and put $q = \exp(-(w - \ell)/2 + \eta_N \hat{v}_N)$. If $q + N^{-1/3 + \varepsilon} \geq 1$, the bound $d_{TV} \leq 1$ suffices. Otherwise $q < 1$ implies $\ell \leq w$, the second exponential is at most q , and $K \leq e^w \leq N^{\varepsilon/2}$ for all sufficiently large N . This proves (4.9). In particular intensities below one are covered directly, without switching to a different conditioning argument. All thresholds precede the length in the fixed band; the fibrewise proof gives the claimed mean distance for the full $\mathcal{F}_{\hat{Y}_N}$. $\square$

The hard estimate is more accurate at bounded intensity. The soft estimate permits larger intensities: $\log^+ \lambda_N \leq \hat{V}_N - c\hat{v}_N$ for fixed $c > 0$ suffices. In particular the hard-cutoff endpoint $\log \lambda_N = V_N + O(1)$ is covered, but no convergence at the exact new endpoint $\log \lambda_N = \hat{V}_N$ is inferred.

Write $(z)_2 = z(z - 1)$ for the second falling factorial.

Corollary 4.4 (First two moments). *At critical lengths,*

$$\begin{aligned} \mathbb{E}Z_{N,L} &= \lambda_N + O(N^{-1/2 + o(1)}), \\ \mathbb{E}(Z_{N,L})_2 &= \lambda_N^2 + O_\varepsilon(N^{-1/3 + \varepsilon}), \\ \text{Var } Z_{N,L} &= \lambda_N + O_\varepsilon(N^{-1/3 + \varepsilon}). \end{aligned}$$

Proof. The first assertion is Corollary 2.5. Expand the second factorial moment over ordered distinct starts. Overlaps cost $O(NLp^2)$; touching pairs cost $p^2N^{1+o(1)}$ by Lemma 2.8. Separated pairs differ from p^2 by at most $p^2(2^{\rho(x,y)} - 1)$, whose sum is $O_\varepsilon(N^{-1/3 + \varepsilon})$. The variance identity concludes. This argument proves the moments directly: total variation alone would not imply convergence of unbounded moments. $\square$

5 Rare word fields, exact marks and the Poisson staircase

The two-window estimate controls dependence between distant words. Their local dependence has a different origin: a suffix of one word may coincide with a prefix of another. Keeping these two issues separate gives a joint law for a growing dictionary of prescribed words, and then a sign-resolved law for long runs.

5.1 A dictionary of prescribed words

Joint word and clump approximations are classical in Markov sequences [28]. Here the local overlap condition is kept separate from the arithmetic estimate for separated windows. Let $\mathcal{W} \subset \{\pm 1\}^B$ be a nonempty deterministic set of distinct words, and write $m = |\mathcal{W}|$, $p_B = 2^{-B}$

and $\Lambda = Nmp_B$. All signs, including the first one, are prescribed. Define

$$I_{x,w} = \mathbf{1}_{\{f(x-1+j)=w_j \ (0 \leq j < B)\}}, \quad x \in I_N, \quad w \in \mathcal{W}.$$

The first letter is at $x - 1$; this indexing matches the left-border convention for runs. For $1 \leq d < B$ write $w \sim_d v$ when $w_{d+j} = v_j$ for $0 \leq j < B - d$. The directed overlap weight is

$$\Omega(\mathcal{W}) = \frac{1}{m} \sum_{w,v \in \mathcal{W}} \sum_{d=1}^{B-1} 2^{-d} \mathbf{1}_{\{w \sim_d v\}}. \quad (5.1)$$

The normalization makes $\Lambda\Omega$ the expected scale of ordered local pairs, up to an absolute factor. It includes self-overlaps and cross-overlaps; controlling each word separately would miss the latter. For example, $(+, +, -)$ followed one site later by $(+, -, +)$ is compatible: the two occurrences fit into $(+, +, -, +)$. The overlap is a constraint on the words themselves, before the values of the multiplicative function are considered.

Theorem 5.1 (Joint Poisson approximation for a word dictionary). *Fix $0 < \beta_- < \beta_+ < \infty$ and $0 < \varepsilon < 1/3$. Uniformly for $\beta_- \log N \leq B \leq \beta_+ \log N$ and all dictionaries $\mathcal{W}$,*

$$\begin{aligned} \mathbb{E} d_{\text{TV}} \left(\mathcal{L}((I_{x,w})_{x \in I_N, w \in \mathcal{W}} \mid \mathcal{F}_{Y_N^\#}), \bigotimes_{x \in I_N, w \in \mathcal{W}} \text{Pois}(p_B) \right) \\ \ll_{\beta_-, \beta_+, \varepsilon} \Lambda\Omega(\mathcal{W}) + \Lambda(1 + \Lambda)e^{-V_N + o(\nu_N)} \\ + N^{-1/3+\varepsilon} \left(\Lambda + \Lambda^{11/6} m^{1/6} + \Lambda^{4/3} m^{2/3} \right). \end{aligned} \quad (5.2)$$

The implied constant and the little-oh term are independent of the words and of m . In particular, fix $0 < \eta < 1/2$ and $C > 0$. If

$$m_N \leq N^{1/2-\eta}, \quad |B_N - \log_2(Nm_N)| \leq C, \quad \Omega(\mathcal{W}_N) \longrightarrow 0, \quad (5.3)$$

then the entire site-and-word-labelled field converges in total variation to the independent Poisson field. Its error is

$$O_{C,\eta} \left(\Omega(\mathcal{W}_N) + e^{-V_N + o(\nu_N)} + N^{-\eta/2} \right). \quad (5.4)$$

Every deterministic restriction of the site-word index set, and every measurable function of the field, satisfies the same bound.

Proof. Use $Y = Y_N^\#$ and remove starts whose B -vertex support contains a Y -defective integer. Let D_Y^{val} denote this set and put $\mathcal{M}_B^{\text{val}} = \sum_{x \in I_N} (2^{m_B(x)} - 1)$ and $a = mp_B$. Distinct words at a site are mutually exclusive, so $0 < a \leq 1$ and $\Lambda = Na$. By Corollary 2.6, deleting the actual and target coordinates costs at most

$$a\mathcal{M}_B^{\text{val}} + 2a|D_Y^{\text{val}}|.$$

All retained word events have conditional marginal p_B . Join all words at the same site and all words at sites with intersecting prime supports above Y , adding edges for overlapping supports if necessary. After conditioning this is an exact dependency graph. Its first Poisson term is $O(a^2(NB + E_Y))$, where E_Y counts ordered pairs of distinct sites sharing a prime above Y .

For $y = x + d$ with $1 \leq d < B$, two specified words are incompatible unless $w \sim_d v$. In the compatible case they prescribe all $B + d$ values in the union. Every vertex of a good union has

a private coordinate above Y , because its diameter is less than $2B < Y$. Hence

$$\mathbb{P}(I_{x,w}I_{x+d,v} = 1 \mid \mathcal{F}_Y) = 2^{-(B+d)} \mathbf{1}_{\{w \sim_d v\}}. \quad (5.5)$$

Summing ordered local pairs costs at most $2\Lambda\Omega(\mathcal{W})$. Different words at the same site contribute zero to the second term.

For separated good sites put $I_x^{\mathcal{W}} = \sum_{w \in \mathcal{W}} I_{x,w}$. This is an indicator, not a count with possible multiplicities, and $\mathbb{P}(I_x^{\mathcal{W}} = 1) = a$. Fourier inversion on the valuation rows also gives $\mathbb{P}(I_x^{\mathcal{W}} I_y^{\mathcal{W}} = 1) \leq a^2 2^{\rho_B^{\text{val}}(x,y)}$. Combining the two bounds *before* summing over sites yields

$$\begin{aligned} \sum_{w,v \in \mathcal{W}} \mathbb{P}(I_{x,w} I_{y,v} = 1) &= \mathbb{P}(I_x^{\mathcal{W}} I_y^{\mathcal{W}} = 1) \\ &\leq \min\{a, a^2 2^{\rho_B^{\text{val}}(x,y)}\} \\ &\leq a^2 \left(1 + \min\{a^{-1}, 2^{\rho_B^{\text{val}}(x,y)} - 1\}\right). \end{aligned} \quad (5.6)$$

Only the error summands have been grouped: the Poisson theorem is still applied to the full family indexed by (x, w) . No word or position label is discarded. Its averaged second term over separated graph edges is therefore at most $a^2(E_Y + R_2^{\text{val}, [1/a]}(N, B))$. We have proved the finite transfer inequality

$$\mathbb{E} d_{\text{TV}} \ll a(\mathcal{M}_B^{\text{val}} + |D_Y^{\text{val}}|) + a^2 \left(NB + E_Y + R_2^{\text{val}, [1/a]}(N, B) \right) + \Lambda\Omega(\mathcal{W}). \quad (5.7)$$

Here $\mathcal{M}_B^{\text{val}} \leq N^{1/2+o(1)}$, $|D_Y^{\text{val}}|/N \leq e^{-V_N+o(\nu_N)}$, and $E_Y/N^2 \leq e^{-V_N+o(\nu_N)} + N^{-1+o(1)}$. Use proposition 3.25 with $T = N/\Lambda$ and $Q_B = 2^B = Nm/\Lambda \geq T$. Its three contributions, after multiplication by a^2 , are bounded by

$$N^{-1/3+\varepsilon} \left(\Lambda^{11/6} m^{1/6} + \Lambda^{4/3} m^{2/3} + \Lambda \right).$$

The smaller polynomial errors are absorbed with a smaller intermediate ε : for example, $m \geq \Lambda/N$ implies $\Lambda^2/N \leq N^{-1/3} \Lambda^{4/3} m^{2/3}$. This proves (5.2). Under (5.3), Λ is bounded above and below and $B/\log N$ stays in one fixed compact interval. The polynomial term is $O(N^{-\eta/2})$ on taking $\varepsilon = \eta/6$. Restriction and measurable contraction cannot increase total variation. $\square$

The exponent $1/2$ is a sufficient bound, not an optimality claim. The remaining restriction comes from the nonterminal term $NQ_B^{2/3}$: after normalization it gives $N^{-1/3+\varepsilon} m^{2/3}$ at bounded intensity. The terminal term has instead been capped by the marginal probability. This distinction matters: a large raw relation count need not produce a proportionally large probability of seeing a dictionary word.

Corollary 5.2 (Replacement by independent signs). *Let (ξ_n) be independent fair signs and let $I_{x,w}^{\text{iid}}$ be the same word indicators formed from them. Under (5.3),*

$$d_{\text{TV}} \left(\mathcal{L}((I_{x,w})_{x,w}), \mathcal{L}((I_{x,w}^{\text{iid}})_{x,w}) \right) \longrightarrow 0.$$

In particular the vector of counts of the individual words is jointly close to independent Poisson variables of mean $N2^{-B}$, even when the number of words tends to infinity.

Proof. For the independent sequence the graph needs only overlapping supports. Equation (5.5) is exact without conditioning, and its process Poisson error is $O(\Lambda\Omega(\mathcal{W}) + \Lambda^2 B/N)$. Compare both fields with the same independent Poisson field and apply Theorem 5.1. Summing sites is a measurable contraction. $\square$

Corollary 5.3 (Most dictionaries have small overlap). *Let $B \geq 1$ and $1 \leq m \leq 2^B$. Choose an m -element subset $\mathcal{W}$ uniformly among the subsets of $\{\pm 1\}^B$. This auxiliary choice is independent of f . Then*

$$\mathbb{E}_{\mathcal{W}} \Omega(\mathcal{W}) = \frac{m(B-1)}{2^B}. \quad (5.8)$$

Under the first two conditions in (5.3), the fraction of dictionaries with $\Omega(\mathcal{W}) > N^{-1/2}$ is $O_{C,\eta}(\log N / \sqrt{N})$. Every dictionary outside this exceptional set has the field approximation (5.4).

Proof. Put $Q = 2^B$. At each proper displacement $1 \leq d < B$, there are 2^d self-compatible words and 2^{B+d} ordered compatible pairs. The number of distinct compatible pairs is therefore $2^d(Q-1)$. A word belongs to the dictionary with probability m/Q ; a distinct ordered pair belongs with probability $m(m-1)/(Q(Q-1))$. After multiplication by $2^{-d}/m$, the expected contribution of this displacement is exactly

$$\frac{2^{-d}}{m} \left(2^d \frac{m}{Q} + 2^d(Q-1) \frac{m(m-1)}{Q(Q-1)} \right) = \frac{m}{Q}.$$

Sum over the $B-1$ displacements to get (5.8). This also covers $m=1$; for $B=1$ the sum is empty. At criticality $m2^{-B} = \Lambda/N = O_C(1/N)$ and $B = O_\eta(\log N)$. Markov's inequality proves the exceptional-fraction estimate. The result is a statement about the size of a class of deterministic dictionaries; no additional randomness is required in Theorem 5.1. $\square$

Corollary 5.4 (Explicit dictionaries with no overlaps). *For all sufficiently large B , there is an explicitly described family $\mathcal{C}_B \subset \{0,1\}^B$ of size at least $2^B/(32B)$ with no proper suffix of any word equal to a prefix of any word, including itself. Every subset of $\mathcal{C}_B$, after identifying 0,1 with $-1,+1$, has $\Omega = 0$. In particular deterministic examples exist throughout the range (5.3).*

Proof. This is the binary marker construction of Chee, Kiah, Purkayastha and Wang [8, Section III]; we include the short argument and the coarse size bound needed here. Set $k = \lceil \log_2(2B) \rceil$ and take all words

$$0^k 1 u 1, \quad |u| = B - k - 2, \quad u \text{ contains no } 0^k.$$

The only occurrence of 0^k in such a word starts at its first position: the first following letter and the last letter are both 1. An overlap of length at most k is impossible because the prefix is all zeros and the suffix ends in 1. A longer proper overlap would put 0^k at a later position, also impossible. There are 2^{B-k-2} choices before the restriction on u . A union bound over at most B positions excludes at most the fraction $B2^{-k} \leq 1/2$. Thus $|\mathcal{C}_B| \geq 2^{B-k-3} \geq 2^B/(32B)$. At the critical dictionary length, $m/2^B = O(1/N)$ and $B = O(\log N)$, so any required m words can be selected from this family for large N . $\square$

For example, one may select $\lfloor N^{2/5} \rfloor$ such words at length $B = (7/5) \log_2 N + O(1)$. Their full labelled field has a Poisson approximation. The coding construction is classical; the conclusion for the multiplicative sequence is the application of the theorem above.

5.2 Words up to a common sign

For $a = (a_0, \dots, a_L) \in \{\pm 1\}^{L+1}$ put $I_x(a) = I_{x,a} + I_{x,-a}$. Define

$$\mathcal{O}(a) = \{1 \leq d \leq L : \exists \eta \in \{\pm 1\}, a_{d+j} = \eta a_j \ (0 \leq j \leq L-d)\}, \quad \Theta(a) = \sum_{d \in \mathcal{O}(a)} 2^{-d}. \quad (5.9)$$

For the two-word dictionary $\{a, -a\}$ one has $\Omega(\{a, -a\}) = \Theta(a)$: every compatible shift has two directed sign choices.

Corollary 5.5 (Poisson law for low-overlap patterns). *Fix $C > 0$ and $\varepsilon > 0$. Uniformly in $a = a_N$ and $|L - \log_2 N| \leq C$,*

$$d_{\text{TV}} \left(\mathcal{L} \left(\sum_{x \in I_N} I_x(a) \right), \text{Pois}(N2^{-L}) \right) \ll_{C,\varepsilon} \Theta(a) + e^{-V_N + o_C(\nu_N)} + N^{-1/3+\varepsilon}. \quad (5.10)$$

The same bound holds under any deterministic spatial mask, with target mean $|A_N|2^{-L}$.

Proof. Apply Theorem 5.1 with $B = L + 1$, $m = 2$, and merge the two word coordinates. Their events are disjoint and their target means add. $\square$

If the least compatible shift $d_*(a_N)$ tends to infinity, then $\Theta(a_N) \leq 2^{1-d_*(a_N)} \rightarrow 0$. For the run-start word $(-1, +1, \dots, +1)$ only $d = L$ is compatible. Periodic words can instead retain a short overlap and need declustering. Notice that the preceding dictionary theorem also separates a and $-a$: their asymptotic independence is not assumed in this corollary.

The dictionary comparison now controls the rare occurrences of each prescribed word. We return to constant runs and keep their attributes together: a start, its full length, its sign and all thresholds it exceeds.

5.3 Exact marks without a growing-mark loss

Almost surely every rightward constant run is finite. Indeed constancy on $[x, \infty)$ would make all prime signs beyond x equal, an event of probability zero; take a countable union over x . Thus

$$J_{x,L} = \sum_{e \geq 0} K_{x,e}, \quad K_{x,e} = \mathbf{1}_{\{f(x-1)=-f(x), f(x)=\dots=f(x+L+e-1), f(x+L+e)=-f(x)\}}. \quad (5.11)$$

This system has $L + e + 1$ independent homogeneous rows on a good support and conditional marginal $p_e = 2^{-L-e-1}$. Its homogeneous rows are nested in those for a larger mark, although the terminal affine right-hand side changes. Marks at the same start are mutually exclusive. We may retain the sign as well:

$$K_{x,e}^s = K_{x,e} \mathbf{1}_{\{f(x)=s\}}, \quad p_{e,s} = 2^{-L-e-2}, \quad s \in \{\pm 1\}. \quad (5.12)$$

On a good support this event prescribes every value, so its conditional probability is $p_{e,s}$. The exact identities are $\sum_s K_{x,e}^s = K_{x,e}$ and $\sum_{e,s} K_{x,e}^s = J_{x,L}$.

Theorem 5.6 (Finite exact-mark comparison). *Let $E \geq 0$, $Q = L + E + 1$, and assume $\beta_- \log N \leq L + 1 \leq Q + 1 \leq \beta_+ \log N$. For $Y > 2(Q + 1)$ let $D_Y(Q)$ be the starts whose support $\{x - 1, \dots, x + Q - 1\}$ contains a Y -defective integer, and let $E_Y(Q)$ be the ordered graph-edge count for these maximal supports. Then*

$$\mathbb{E} d_{\text{TV}} \left(\mathcal{L}((K_{x,e})_{x \in I_N, 0 \leq e \leq E} \mid \mathcal{F}_Y), \bigotimes_{x \in I_N, 0 \leq e \leq E} \text{Pois}(p_e) \right) \ll p \mathcal{M}_B + p |D_Y(Q)| + p^2 (NQ + E_Y(Q) + R_2(N, L)), \quad p = 2^{-L}. \quad (5.13)$$

Exactly the same upper bound holds for the sign-resolved family $(K_{x,e}^s)_{x \in I_N, 0 \leq e \leq E, s \in \{\pm 1\}}$, compared with independent Poisson variables of means $p_{e,s}$. There is no polynomial factor in the

number of marks. At $Y = Y_N^\sharp$, the right side is at most

$$C_\varepsilon \lambda_N (1 + \lambda_N) \left(e^{-V_N + o(\nu_N)} + N^{-1/3+\varepsilon} \right). \quad (5.14)$$

Proof. On good maximal supports all marginals are p_e . Apply the process Poisson theorem to the marked family, allowing all marks on neighbouring supports to be neighbours, and all marks at the same site to be neighbours. The two identities

$$\sum_{e=0}^E p_e \leq p, \quad \sum_{e,f=0}^E \mathbb{P}(K_{x,e} K_{y,f} = 1) \leq \mathbb{P}(J_{x,L} J_{y,L} = 1) \quad (5.15)$$

remove any double sum over marks before estimating the pair mass. Local pairs cost $O(NQp^2)$ and separated pairs cost $p^2(E_Y(Q) + R_2(N, L))$. Bad actual marks at one site have total mass at most that of $J_{x,L}$, so their deletion is bounded by $p\mathcal{M}_B + p|D_Y(Q)|$, regardless of which late vertex caused removal. This proves (5.13). The bounds $|D_Y(Q)| \leq (Q+1)A(3N, Y)$ and (4.2) with B replaced by $Q+1$, together with Theorem 3.1 and proposition 4.2, give (5.14).

For the signed version, the same-site alternatives are still disjoint and both sums in (5.15) remain valid after summing signs. It remains to check local pairs before that summation. Put $a = L + e$, $b = L + f$, and $d = y - x > 0$. If $d < a$ the pair is impossible. If $d = a$, the two prescribed value strings share two vertices, require the second run sign to be the opposite of the first, and have probability $4p_{e,s}p_{f,t}$ when compatible. If $d = a + 1$, they share one vertex, require equal run signs, and have probability $2p_{e,s}p_{f,t}$. For $d > a + 1$ their actual supports are disjoint, and a good local union has probability $p_{e,s}p_{f,t}$. These assertions follow by counting the prescribed values in a union whose private coordinates make them uniform. Thus the local sum is still $O(NQp^2)$ and all other terms are unchanged. $\square$

The use of (5.15) is essential. An exact mark $e < E$ does not imply a threshold event at length $L + E + 1$; a bad vertex can occur beyond its support. We bound deletion by the base event, not the longest event.

The exact tail identity is

$$\{\exists x \in I_N, e > E : K_{x,e} = 1\} = \{Z_{N,L+E+1} > 0\}. \quad (5.16)$$

The first-moment estimate bounds this probability by $\lambda_N 2^{-(E+1)}(1 + N^{-1/2+o(1)})$ whenever the enlarged length is in a fixed logarithmic band. The independent target tail has mass $\lambda_N 2^{-(E+1)}$. Couple both tails to zero. At criticality choose $E = \lceil V_N / \log 2 \rceil$. This is $o(\log N)$; (5.14) and (5.16) prove the full-field estimate, including signs, in Theorem 1.1. Riemann sums and geometric tail tightness give its weak diffuse limit.

Corollary 5.7 (An explicit compound-Poisson cluster law). *Let $C_{N,L}$ count all constant length- L windows whose left endpoint lies in I_N , without requiring left maximality. At criticality its law is at distance $o(1)$ from that of*

$$\sum_{i=1}^{P_N} H_i, \quad P_N \sim \text{Pois}(\lambda_N), \quad \mathbb{P}(H_i = h) = 2^{-h}, \quad h \geq 1,$$

with independent H_i and P_N . The target probability generating function is $\exp[\lambda_N(z/(2-z)-1)]$ for $0 \leq z \leq 1$.

Proof. A maximal run of length $L + e$ contributes $e + 1$ windows. Thus, apart from runs crossing an endpoint of I_N , $C_{N,L} = \sum_{x \in I_N, e \geq 0} (e + 1) K_{x,e}$. A discrepancy implies a constant length- L

window starting at N or at $2N - 1$. Each such event has $L - 1$ relative constraints; private pivots and the pointwise defect bound give probability at most $2^{-(L-1)+O(\log N/\log \log N)} = N^{-1+o(1)}$. The full marked-field comparison contracts under the measurable weighted sum. Its target is the stated compound Poisson law, since $H = e + 1$. No moment convergence is inferred from this contraction. $\square$

5.4 Growing depth: distinguish positions from counts

Put $b_N = \lfloor \log_2 N \rfloor$, $\vartheta_N = \log_2 N - b_N$. For $d_N \in \mathbb{N}_0$ with $d_N = o(\log N)$, set

$$L_N = b_N - d_N, \quad \Lambda_N = N2^{-L_N} = 2^{\vartheta_N + d_N}.$$

Let $K_{x,q}^{\text{ex}}$ denote the exact-length event for a left-maximal run of length q (zero for $q < 1$). The moving field and its target are

$$\mathcal{E}_{N,d_N} = \sum_{x \in I_N} \sum_{r \geq -d_N} K_{x,b_N+r}^{\text{ex}} \delta_{(x/N,r)}, \quad \mathbb{E} \Pi_{N,d_N} \{(x/N,r)\} = 2^{-b_N-r-1}, \quad (5.17)$$

where the target atoms are independent Poisson variables. Their total mean is Λ_N . Define exact-level and threshold counts by

$$X_{N,r} = \sum_{x \in I_N} K_{x,b_N+r}^{\text{ex}}, \quad T_{N,m} = \sum_{r \geq m} X_{N,r} = Z_{N,b_N+m}.$$

For the target write $P_{N,r} = \Pi_{N,d_N}([1,2] \times \{r\})$ and $S_{N,m} = \sum_{r \geq m} P_{N,r}$. Thus $P_{N,r} \sim \text{Pois}(2^{\vartheta_N-r-1})$ independently.

The preceding estimates bound the mean, over the small-prime environment, of a conditional total-variation distance. Restricting this average to an event $C_N \in \mathcal{F}_Y$ multiplies the error by at most $1/\mathbb{P}(C_N)$; this is the stable-kernel principle of Lemma 6.1. It is the first information cost below.

Theorem 5.8 (Two levels of joint approximation). *Fix $c > 0$ and $0 < \varepsilon < 1/3$. Let C_N be an arithmetic conditioning event of positive probability and put $I_N^{\text{info}} = -\log \mathbb{P}(C_N)$.*

(i) *Labelled field. If $C_N \in \mathcal{F}_{Y_N^\sharp}$, a sufficient condition for full-field total variation to tend to zero is*

$$I_N^{\text{info}} + \log \Lambda_N + \log(1 + \Lambda_N) \leq V_N - c\nu_N. \quad (5.18)$$

At a movable cutoff, if $C_N \in \mathcal{F}_{\widehat{Y}_N}$, the sufficient condition improves to

$$I_N^{\text{info}} + \log \Lambda_N \leq \widehat{V}_N/2 - c\widehat{\nu}_N. \quad (5.19)$$

In the nontrivial range, the latter comparison has upper bound

$$\begin{aligned} & e^{I_N^{\text{info}} + \log \Lambda_N - \widehat{V}_N/2 + o(\widehat{\nu}_N)} + e^{I_N^{\text{info}} + 2 \log \Lambda_N - \widehat{V}_N + o(\widehat{\nu}_N)} \\ & + e^{I_N^{\text{info}} \Lambda_N (1 + \Lambda_N) N^{-1/3+\varepsilon+o(1)}}. \end{aligned} \quad (5.20)$$

(ii) *Aggregated exact levels. If $C_N \in \mathcal{F}_{Y_N^\sharp}$ and*

$$I_N^{\text{info}} + \log \Lambda_N \leq V_N - c\nu_N, \quad (5.21)$$

then, for every $0 < c' < c$ and $0 < \varepsilon < 1/3$,

$$d_{\text{TV}} \left(\mathcal{L}((X_{N,r})_{r \geq -d_N} \mid C_N), \bigotimes_{r \geq -d_N} \text{Pois}(2^{\vartheta_N - r - 1}) \right) \ll e^{-c' \nu_N} + N^{-1/3 + \varepsilon}. \quad (5.22)$$

Exactly the same distance holds for the full threshold paths $(T_{N,m})_{m \geq -d_N}$ and $(S_{N,m})_{m \geq -d_N}$.

Proof. Conditioning multiplies any averaged kernel error by at most $e^{I_N^{\text{info}}}$. For (i), apply Theorem 5.6 with base L_N and then remove the mark tail using (5.16). For the hard bound one may take $E = \lceil (I_N^{\text{info}} + V_N + \log(2 + \Lambda_N)) / \log 2 \rceil$. Under (5.18) this is $o(\log N)$ and both tail costs are absorbed by the displayed error. At a free cutoff $Y = e^w$, the leading process errors are

$$e^{I_N^{\text{info}}} \Lambda_N e^{-D(H_N/w) + o(H_N/w)}, \quad e^{I_N^{\text{info}}} \Lambda_N^2 e^{-w + o(H_N/w)}.$$

Choose $w = \hat{V}_N$, so $D(H_N/w) = w/2$, and take an equally large $o(\log N)$ tail cutoff. This proves (5.20) and (5.19). The hard choice remains sharper at bounded intensity; the movable choice improves the admissible growing intensity.

For (ii), aggregate sites *before* solving the multivariate Poisson Stein equation. If $q_e = 2^{-(e+1)}$ are the normalized category masses, Barbour's Lemmas 2–3 [5, p. 179], as applied in the companion, bound the directional second-difference factor by

$$C \frac{1 + \log^+(2\Lambda_N)}{\Lambda_N \sqrt{q_e q_f}}.$$

The mark sum remains uniform because $\sum_{e \geq 0} \sqrt{q_e} = 1 + \sqrt{2}$. The full argument and the precise finite-cutoff bound are given in Theorem C.1. It yields a leading error $e^{I_N^{\text{info}}} \Lambda_N (1 + \log^+(2\Lambda_N)) e^{-V_N + o(\nu_N)}$ and a polynomial remainder after the enlarged-length arithmetic estimate. All subpolynomial factors in that remainder are absorbed by choosing a smaller initial ε . Choose a constant strictly between c' and c for the tail cutoff. The scalar relative estimate at the shifted length bounds both tails by $O(e^{-c' \nu_N})$; see Theorem C.1. Finally, on finite-mass sequences, the map $X_r \mapsto \sum_{s \geq r} X_s$ is a measurable bijection with inverse $X_r = T_r - T_{r+1}$. Total variation is preserved, not merely contracted. $\square$

Part (ii) retains every statistic of the multiset of lengths, but not the positions. A maximal coupling identifies all those retained statistics at once, with no union bound over thresholds. A field coupling from (i) also retains positions. The ranges are sufficient bounds for these methods, not phase transitions. In particular the full one-factor labelled range is not proved by these comparisons.

Corollary 5.9 (Sign-resolved growing fields). *Both conclusions of Theorem 5.8 remain valid after retaining the sign of each run. The labelled target has atom means $2^{-b_N - r - 2}$ for each $s \in \{\pm 1\}$, and the aggregated target has independent means $2^{\vartheta_N - r - 2}$ at each pair (r, s) .*

Proof. For the labelled result use the signed version of Theorem 5.6 and the unchanged tail identity. For aggregation, the normalized category weights are $q_{e,s} = 2^{-e-2}$ and $\sum_{e,s} \sqrt{q_{e,s}} = \sqrt{2}(1 + \sqrt{2})$. The directional Stein bound therefore has the same summability. At the largest retained support, the full value-relation space dominates each mixed signed kernel by extension of coefficient vectors by zero; proposition 3.24 supplies the same two-window envelope. The retained support has $L + E + 2$ vertices, differing by only one from the relative-mark convention. Since $E = O(V_N) = o(\log N)$, its polynomial remainder is absorbed exactly as in part (ii). The signed local-pair calculation and the same tail bound finish the proof. $\square$

5.5 Target kernels, extremes and the Gaussian lower edge

The independent target levels give, exactly,

$$S_{N,m} \sim \text{Pois}(2^{\vartheta_N - m}), \quad (5.23)$$

$$S_{N,m+j} \mid S_{N,m} = n \sim \text{Bin}(n, 2^{-j}), \quad (5.24)$$

$$S_{N,m-j} \mid S_{N,m} = n \stackrel{d}{=} n + \text{Pois}(2^{\vartheta_N - m}(2^j - 1)), \quad (5.25)$$

when the lower levels are retained. The whole forward path is the number of survivors among n independent geometric lifetimes, each with $\mathbb{P}(E = e) = 2^{-(e+1)}$. These are identities for the Poisson target, not exact finite- N Markov assertions for the multiplicative sequence.

For any integer $m \geq 0$, the dyadic excess maximum $M_{N,L} = \max\{\ell_x - L : x \in I_N, J_{x,L} = 1\}$, with empty value $-\infty$, satisfies the exact identity

$$\{M_{N,L} \leq m\} = \{Z_{N,L+m+1} = 0\}. \quad (5.26)$$

Thus the scalar estimate at $L' = L + m + 1$, with $\mu = N2^{-L'}$, gives

$$\mathbb{P}(M_{N,L} \leq m) = e^{-\mu} + O\left(\min\{1, \mu(e^{-V_N + o(\nu_N)} + N^{-1/3+\varepsilon})\}\right)$$

whenever $L' + 1$ is in a fixed logarithmic band. In particular one-hit upper tails are relative when $N2^{-L'} \rightarrow 0$. For the k th largest run length among starts in I_N , along a subsequence $\vartheta_N \rightarrow \vartheta$,

$$\mathbb{P}(R_N^{[k]} - b_N < m) \rightarrow e^{-2^{\vartheta-m}} \sum_{j=0}^{k-1} \frac{(2^{\vartheta-m})^j}{j!}. \quad (5.27)$$

This is just the event that fewer than k starts exceed the threshold.

Theorem 5.10 (Joint Poisson–Gaussian limit). *Assume $d_N \rightarrow \infty$ and the hypotheses of part (ii) of Theorem 5.8. Put*

$$G_N(j) = \frac{Z_{N,b_N-d_N+j} - \Lambda_N 2^{-j}}{\sqrt{\Lambda_N}}.$$

Along $\vartheta_N \rightarrow \vartheta$, for every fixed J and every finite $R \subset \mathbb{Z}$, conditionally on C_N ,

$$((G_N(j))_{0 \leq j \leq J}, (X_{N,r})_{r \in R}) \implies ((G(j))_{0 \leq j \leq J}, (P_r^{(\vartheta)})_{r \in R}),$$

where $\mathbb{E}G(j)G(k) = 2^{-\max(j,k)}$, the variables $P_r^{(\vartheta)} \sim \text{Pois}(2^{\vartheta-r-1})$ are independent, and the Gaussian and Poisson vectors are independent. The normalized Gaussian sequence $H(j) = 2^{j/2}G(j)$ is the stationary AR(1) chain $H(j+1) = 2^{-1/2}H(j) + 2^{-1/2}\zeta_{j+1}$, where $(\zeta_j)_{j \geq 1}$ are independent standard normal random variables, independent of $H(0) \sim N(0,1)$.

Proof. Transfer to the independent target by (5.22). Choose a fixed $r_0 < \min R$ (any fixed r_0 if R is empty). The lower-edge sums restricted to $r < r_0$ are independent of the displayed critical levels and satisfy the multivariate Poisson central limit theorem. Their covariance after normalization tends to $2^{-\max(j,k)}$. The centered sum over $r \geq r_0$ has bounded variance and hence vanishes after division by $\sqrt{\Lambda_N}$. This proves the joint limit and independence. The covariance of H is $2^{-|j-k|/2}$, giving the stated Gaussian autoregression. $\square$

For fixed compact sets of levels, the untruncated centered field has the phase-indexed target

$$\text{PPP} \left(dt \otimes \sum_{r \in \mathbb{Z}} 2^{\vartheta_N - r - 1} \delta_r \right).$$

Riemann sums give the corresponding Laplace approximation without assuming that ϑ_N converges. A phase-convergent subsequence turns it into a fixed weak limit; its intensity is locally finite although it diverges towards $r = -\infty$. Full formulas and conditional configuration statements are in [section D.4](#). No continuous run lengths or canonical coupling simultaneously over all N is asserted.

6 Arithmetic information and resolved conditional laws

There are two different conditioning operations. An arithmetic event fixes information about prime signs before the Poisson comparison. Conditioning on a count or another observable then divides by the probability of that observable. Keeping the two costs separate avoids treating a rare state as though an absolute error resolved it automatically.

The following table collects the sufficient budgets. Here $I = -\log \mathbb{P}(C)$, $c > 0$ is fixed, $Y^\# = \lfloor e^V \rfloor$ and $\hat{Y} = \lfloor e^{\hat{V}} \rfloor$; ν and $\hat{\nu}$ belong to those cutoffs. The moving-field intensities Λ are at least one.

Observable	Sufficient budget	Prime field
Scalar, hard	$I + \log^+ \lambda \leq V - c\nu$	$\mathcal{F}_{Y^\#}$
Scalar, soft	$2I + \log^+ \lambda \leq \hat{V} - c\hat{\nu}$	$\mathcal{F}_{\hat{Y}}$
All exact-level counts	$I + \log \Lambda \leq V - c\nu$	$\mathcal{F}_{Y^\#}$
All sites, hard	$I + \log \Lambda + \log(1 + \Lambda) \leq V - c\nu$	$\mathcal{F}_{Y^\#}$
All sites, optimized	$I + \log \Lambda \leq \hat{V}/2 - c\hat{\nu}$	$\mathcal{F}_{\hat{Y}}$

Exact marks and signs can be retained in both field rows. The inequalities have fixed negative margins; equality at the boundary and optimality for the model are not asserted. A further conditioning on an observed count has the separate probability cost analysed below.

6.1 Stable kernels and information cost

Lemma 6.1 (Stable product lift). *Let W take values in a standard Borel space, let $\mathcal{F}$ be a sigma-field, and let ν be a deterministic target law. If*

$$\mathbb{E} d_{\text{TV}}(\mathcal{L}(W \mid \mathcal{F}), \nu) \leq \varepsilon,$$

then for every $\mathcal{F}$ -measurable random element V ,

$$d_{\text{TV}}(\mathcal{L}(V, W), \mathcal{L}(V) \otimes \nu) \leq \varepsilon.$$

For $C \in \mathcal{F}$ with $\mathbb{P}(C) > 0$, the conditional bound is at most $\varepsilon/\mathbb{P}(C)$, both for W and for the joint product lift with V .

Proof. Disintegrate the actual and product laws over $\mathcal{F}$. For a test set, the difference of its conditional sections is bounded by the conditional total-variation distance. Integrate, or integrate over C and divide by $\mathbb{P}(C)$. Countable configuration spaces used here have measurable kernels; the standard-Borel formulation permits the additional recorded variables. $\square$

For $C_N \in \mathcal{F}_{Y_N^\#}$ write $I_N^{\text{info}} = -\log \mathbb{P}(C_N)$. The hard scalar estimate becomes

$$d_{\text{TV}}(\mathcal{L}(Z_{N,L} \mid C_N), \text{Pois}(\lambda_N)) \ll \min\{1, e^{I_N^{\text{info}}} \lambda_N (e^{-V_N + o(\nu_N)} + N^{-1/3+\varepsilon})\}. \quad (6.1)$$

For $C_N \in \mathcal{F}_{\widehat{Y}_N}$ the soft proof gives instead

$$d_{\text{TV}}(\mathcal{L}(Z_{N,L} \mid C_N), \text{Pois}(\lambda_N)) \ll \min\{1, e^{I_N^{\text{info}} - (\widehat{V}_N - \log^+ \lambda_N)/2 + o(\widehat{\nu}_N)} + e^{I_N^{\text{info}} N^{-1/3 + \varepsilon + o(1)}}\}. \quad (6.2)$$

Thus the sufficient scalar budgets are

$$I_N^{\text{info}} + \log^+ \lambda_N \leq V_N - c\nu_N, \quad 2I_N^{\text{info}} + \log^+ \lambda_N \leq \widehat{V}_N - c\widehat{\nu}_N.$$

The hard route can be preferable at a larger information cost, but it requires measurability with respect to its smaller prime field. These budgets measure probabilities of events, not the number of primes exposed. For instance, specifying k independent prime signs costs $k \log 2$; a rank- k compatible affine cylinder has the same cost.

At small intensity the hard estimate is relative: under $I_N^{\text{info}} \leq V_N - c\nu_N$ and $\lambda_N \rightarrow 0$,

$$\mathbb{P}(Z_{N,L} > 0 \mid C_N) = \lambda_N(1 + o(1)), \quad \mathbb{P}(Z_{N,L} = 1 \mid C_N) = \lambda_N(1 + o(1)). \quad (6.3)$$

The polynomial terms are negligible since $I_N^{\text{info}} = o(\log N)$. At large intensity, either valid scalar budget gives a conditional Gaussian limit after centering by λ_N and dividing by $\sqrt{\lambda_N}$. The target Poisson normal-approximation error must still be added to the arithmetic error; the quantitative and relative forms are in [section D](#).

6.2 The exact resolution cost

Lemma 6.2 (Sharp conditioning inequality). *Let μ, ν be probability laws on $E \times F$ with $d_{\text{TV}}(\mu, \nu) \leq \varepsilon$. Let $B \subset E$ have $p = \nu_E(B) > \varepsilon$ and write $q = \mu_E(B)$. Then $q > 0$ and*

$$d_{\text{TV}}(\mu_F(\cdot \mid B), \nu_F(\cdot \mid B)) \leq \frac{\varepsilon}{\max(p, q)} \leq \frac{\varepsilon}{p}. \quad (6.4)$$

Proof. The bound $|p - q| \leq \varepsilon$ gives positivity. First suppose $q \leq p$. For any measurable $D \subset F$ put $A = B \times D$. Since $\mu(A) \leq q$,

$$\frac{\mu(A)}{q} - \frac{\nu(A)}{p} \leq \frac{\mu(A) - \nu(A) + p - q}{p} = \frac{\nu(B \times D^c) - \mu(B \times D^c)}{p} \leq \frac{\varepsilon}{p}.$$

The supremum of the positive differences equals total variation for probability laws. If $q \geq p$, interchange the two laws. $\square$

The denominator expresses the correct scale: convergence after conditioning requires $\varepsilon/p \rightarrow 0$, not merely $p > \varepsilon$. This lemma applies to an entire future path at once.

Let $\lambda_{N,m} = 2^{\vartheta_{N-m}}$ and $p_{N,m}(n) = e^{-\lambda_{N,m}} \lambda_{N,m}^n / n!$. Write $\varepsilon_{N,m,C}^{\text{path}}$ for the total-variation error between the actual complete future threshold path, under C_N , and the target staircase based at m . The aggregated proof works equally for an arbitrary base threshold in a fixed logarithmic band (including $\lambda_{N,m} < 1$). In its nontrivial information range it gives

$$\varepsilon_{N,m,C}^{\text{path}} \ll e^{I_N^{\text{info}}} \lambda_{N,m} (1 + \log^+(2\lambda_{N,m})) e^{-V_N + o(\nu_N)} + N^{-1/3 + \varepsilon}, \quad (6.5)$$

with an arbitrarily small fixed exponent slack absorbing the enlarged length and the subpolynomial information factor.

Theorem 6.3 (Resolved thinning kernels). *If $p_{N,m}(n) > \varepsilon_{N,m,C}^{\text{path}}$, then*

$$d_{\text{TV}}(\mathcal{L}((Z_{N,b_N+m+j})_{j \geq 0} \mid C_N, Z_{N,b_N+m} = n), \text{Thin}_n) \leq \frac{\varepsilon_{N,m,C}^{\text{path}}}{p_{N,m}(n)}, \quad (6.6)$$

where Thin_n is the path of n independent geometric lifetimes from (5.24). If a fixed lower segment is also retained, the same inequality uses the comparison error based at its lowest level and gives the reverse immigration kernel (5.25) jointly with the future.

Proof. Apply Lemma 6.2 to the present count and the entire retained path. The conditional target is exactly the geometric-lifetime path by independent Poisson splitting. For lower levels, the increments below the present level are independent Poisson variables. A finite intensity half-line comparison cannot control the whole infinite reverse path, whose total target mass diverges. $\square$

For every integer $n \geq 1$ and every $\lambda > 0$, the local formula is

$$p_\lambda(n) = \frac{e^{-\delta_n}}{\sqrt{2\pi n}} e^{-\lambda h(n/\lambda)}, \quad 0 \leq \delta_n \leq \frac{1}{12n}, \quad h(v) = v \log v - v + 1.$$

The correction is independent of λ ; its relative error is at most $1/(12n)$. For $n = 0$ use the exact mass $e^{-\lambda}$ instead. The telescoping proof is in [section D](#). For $n_N \rightarrow \infty$, a sufficient condition for the right side of (6.6) to vanish is

$$I_N^{\text{info}} + \log^+ \lambda_{N,m} + \log(1 + \log^+(2\lambda_{N,m})) + \frac{1}{2} \log n_N + \lambda_{N,m} h(n_N/\lambda_{N,m}) \leq V_N - c\nu_N. \quad (6.7)$$

In the central regime $n_N = \lambda_{N,m} + t_N \sqrt{\lambda_{N,m}} + O(1)$, $|t_N| = o(\lambda_{N,m}^{1/6})$, this reduces, with a fixed margin, to

$$I_N^{\text{info}} + \frac{3}{2} \log \lambda_{N,m} + \frac{1}{2} t_N^2 \leq V_N - c\nu_N.$$

The coefficient $3/2$ is a sufficient error budget, not a threshold of the model. Its derivation, including the common-band check, is in [section D](#).

6.3 Quenched comparison on geometric scales

Corollary 6.4 (Almost-sure environmental control). *Let $N_k \asymp q^k$, $q > 1$, and suppose $\log \Lambda_{N_k} \leq V_{N_k} - c\nu_{N_k}$. For each $0 < \beta < c$, the conditional total-variation distance, given $\mathcal{F}_{Y_{N_k}^\#}$, between the complete aggregated staircase and its deterministic target is at most $e^{-\beta\nu_{N_k}}$ for all sufficiently large k , almost surely.*

Proof. The fibrewise proof of Theorem 5.8(ii) gives mean distance $e^{-c\nu_N + o(\nu_N)} + N^{-1/3 + o(1)}$. Markov's inequality at $e^{-\beta\nu_N}$ gives a summable series along $N_k \asymp q^k$, since $\nu_{N_k} \asymp \sqrt{k/\log k}$. Borel–Cantelli concludes; no independence between scales is used. $\square$

The same argument applies to the labelled field in its own smaller sufficient range, and to scalar kernels uniformly over the $O(\log N)$ lengths of any fixed logarithmic band with a fixed error margin. A maximal coupling exists separately in each qualified environment and scale. This does not furnish a canonical coupling over all scales.

7 Finite prefixes and the arithmetic boundary

A finite prefix is not a union of independent dyadic experiments. Its retained bulk must be treated in one Poisson comparison, using the macroscopic relation estimate. The starts near the origin need a different argument: there the initial run costs one sign per prime, rather than one sign per integer. The two candidate masses are $\alpha_L = 2^{-\pi(L)}$ at the boundary and $\lambda_M = M2^{-L}$ in the bulk. To compare them, we must bound the other microscopic starts, discard the intervening region, and approximate the bulk with error $o(\lambda_M)$. A critical-scale error of size $o(1)$ is not sufficient for this last task.

The limiting source weights arise by normalizing these two masses. The stronger conclusion of Theorem 7.9 also distinguishes their marks: both excess laws are geometric, but one counts new primes and the other new integer positions. This distinction is arithmetic, not a property of the common geometric distribution.

7.1 The exact boundary event

Set

$$\begin{aligned} C_L &= \mathbf{1}_{\{f(1)=\dots=f(L)\}}, & \mathbf{B}_L &= (C_L, (J_{x,L})_{2 \leq x \leq 2L^2}), \\ B_L^{\text{mic}} &= C_L + \sum_{2 \leq x \leq 2L^2} J_{x,L}, & q_L &= \mathbb{P}(B_L^{\text{mic}} > 0). \end{aligned}$$

Because $f(1) = 1$, the event $C_L = 1$ prescribes precisely the prime signs up to L . Hence

$$\mathbb{P}(C_L = 1) = 2^{-\pi(L)} \quad (7.1)$$

exactly. The quantity q_L is different: it includes all other microscopic starts. The next result shows that their combined mass is smaller by a fixed exponential factor in $\pi(L)$.

Theorem 7.1 (Microscopic non-vacancy and localization). *Let $\delta_* = 28271/332640$. As $L \rightarrow \infty$,*

$$\sum_{x=2}^{2L^2} \mathbb{P}(J_{x,L} = 1) \leq 2^{-(1+\delta_*-o(1))\pi(L)}. \quad (7.2)$$

Consequently, for every fixed $0 < \delta_0 < \delta_*$,

$$q_L = 2^{-\pi(L)}(1 + O_{\delta_0}(2^{-\delta_0\pi(L)})). \quad (7.3)$$

In particular $\delta_0 = 1/12$ is admissible. Conditional on microscopic non-vacancy, the exact border is the unique recorded start with probability $1 - O_{\delta_0}(2^{-\delta_0\pi(L)})$.

Proof. Let M_x be the matrix whose rows are $v(n)$, $n \in V_x$. The start matrix is $A_x = D_x M_x$, where

$$D_x z = (z_{x-1} + z_x, z_x + z_{x+1}, \dots, z_x + z_{x+L-1}).$$

The kernel of D_x consists of constant vectors. Therefore

$$\text{rank } A_x \geq \text{rank } M_x - 1, \quad \mathbb{P}(J_{x,L} = 1) \leq 2^{-\text{rank } A_x}. \quad (7.4)$$

If $2 \leq x \leq L-1$, both $2x$ and $2x-2$ lie in the proposed constant run. The first forces $f(2) = 1$, and the second forces $f(2) = -1$. Thus these starts are impossible. For the three transition starts $x \in \{L, L+1, L+2\}$, primes between $2\sqrt{B}$ and B , together with prime vertices above B , give an identity minor of size $(2 - o(1))\pi(B)$. The explicit choice of rows is in [Lemma E.1](#).

For $B + 2 \leq x \leq B^2 + 2$, the uniform Laishram–Shorey bound gives

$$\omega\left(\prod_{n \in V_x} n\right) \geq (2 - o(1))\pi(B).$$

After discarding the prime divisors at most B , at least $t_x \geq (1 - o(1))\pi(B)$ primes $q > B$ remain. All vertices are at most $B^2 + B < q^2$, so such a prime occurs to valuation one at one vertex only; two such primes cannot occupy the same vertex. Their columns span a t_x -dimensional coordinate subspace.

Use the medium primes $B/11 < p \leq B$. Their total odd-valuation incidence count satisfies

$$E_x \geq (H_{11} - 1 - o(1))\pi(B), \quad H_{11} = \sum_{j=1}^{11} j^{-1}. \quad (7.5)$$

Here the loss from even valuations is only $O(1)$ over the entire prime range: if $p^2 \mid n \leq B^2 + B$ then $n = ap^2$ with $a \leq 122$, and for each a there are only $O(1)$ possibilities in an interval of length B . This global count avoids paying one lost incidence per prime.

Delete the private rows. Each deleted row removes at most one medium incidence, each remaining row has weight at most two, each column has weight at most eleven, and a pair of medium primes cannot occur together in two rows. The associated incidence graph is therefore simple. The elementary simple-incidence estimate [Lemma E.2](#) gives rank at least $E/12$. Passing to the quotient by the private coordinate subspace yields

$$\text{rank } M_x \geq t_x + \frac{(E_x - t_x)_+}{12} \geq \left(1 + \frac{H_{11} - 2}{12} - o(1)\right) \pi(B).$$

Since $(H_{11} - 2)/12 = 28271/332640$, (7.4) proves the desired uniform surplus in the quadratic range.

For $x > B^2 + 2$, the pointwise Balasubramanian–Shorey consequence [Lemma E.3](#) gives

$$m_B(x) \leq B - g_L, \quad g_L = \frac{B}{\log B}(\log \log B - \log \log \log B - \theta_0 + o(1)), \quad \frac{g_L}{\pi(B)} \rightarrow \infty. \quad (7.6)$$

The affine bound then gives $\mathbb{P}(J_{x,L} = 1) \leq 2^{-g_L}$. Sum the four ranges; their polynomial number of starts is absorbed in the $o(\pi(B))$ exponent. Since $\pi(B) = \pi(L) + O(1)$, this proves (7.2). Equation (7.1) and a union bound prove the remaining conclusions. The complete incidence count, the uniform literature input and the choice $K = 11$ are detailed in [section E](#), with local product thresholds distinguished from the thresholds of the asymptotic inputs. $\square$

7.2 Discarding deep starts without assuming independence

Lemma 7.2 (Two-defect windows at intermediate heights). *Assume $B \asymp \log M$. The number of starts $x \in [2L^2, 2M] \cap \mathbb{Z}$ with at least two B -defective vertices is at most $\exp(O(\log M / \log \log M))$. The same bound holds for every subpopulation; in particular it is uniform on each $[X, 2X]$ with $2L^2 \leq X \leq M$.*

Proof. Two such vertices have forms $d_1 a^2, d_2 b^2$ differing by a nonzero integer of modulus at most L , with squarefree B -smooth d_i . There are at most $4^{\pi(B)} = \exp(O(\log M / \log \log M))$ coefficient pairs and $O(B^2)$ offset pairs. For $d_1 \neq d_2$, use the uniform Pell bound at ambient scale M ; the coefficients and variables are polynomially bounded in M . For equal coefficients use $(a - b)(a + b) = (i - j)/d_1$ and the divisor bound. A solution determines the start once the

offsets are fixed. This counts the entire population up to height $2M$ in a single Pell box, not once per dyadic interval. $\square$

Proposition 7.3 (Deep-start and mesoscopic estimates). *For each fixed $0 < \delta < 1$ and fixed logarithmic band,*

$$\sum_{2 \leq x < M^\delta} \mathbb{P}(J_{x,L} = 1) \ll \lambda_M M^{\delta-1} + \exp(-c \log M / \log \log M), \quad \lambda_M = M 2^{-L}, \quad (7.7)$$

where $c > 0$ depends only on that band and δ . More precisely,

$$\mathbb{P}(\exists x \in (2L^2, M^\delta) : J_{x,L} = 1) \leq M^\delta 2^{-L} + C_1 e^{C_2 L / \log L} 2^{-g_L}. \quad (7.8)$$

If $L \rightarrow \infty$, $L = O(\log M)$ and $\lambda_M \rightarrow 0$, the latter bound is $o(q_L + \lambda_M)$.

Proof. For any deterministic $A \subset [2L^2, 2M] \cap \mathbb{Z}$, starts with at most one defect have probability at most 2^{-L} . The remaining starts are counted together by Lemma 7.2, and (7.6) bounds each probability by 2^{-g_L} . Uniformly in A ,

$$\sum_{x \in A} \mathbb{P}(J_{x,L} = 1) \leq |A| 2^{-L} + C_1 e^{C_2 L / \log L} 2^{-g_L}.$$

The constants depend only on the fixed logarithmic band. Adding the microscopic range, controlled by Theorem 7.1, proves (7.7); restricting this mask proves (7.8). No exterior factor counting height slices is needed. For the last assertion, $M^\delta 2^{-L} = o(\lambda_M)$, while $g_L - \pi(L) \gg L \log \log L / \log L$ dominates the sieve prefactor. Thus the second term is $o(2^{-\pi(L)}) = o(q_L)$. $\square$

The same argument gives cutoff stability. For $2L^2 \leq K \leq 2^L$,

$$\mathbb{P}(\exists x \in (2L^2, K] : J_{x,L} = 1) \leq K 2^{-L} + O(e^{CL / \log L} 2^{-g_L}). \quad (7.9)$$

One may use the artificial ambient scale 2^L in Lemma 7.2. If $K = o(2^{L-\pi(L)})$, the entire bound is $o(q_L)$. Hence the boundary asymptotic and conditional uniqueness remain valid with cutoff K , and the two conditional microscopic configurations, padded by zeros, are at total-variation distance $o(1)$ by Lemma 6.2.

7.3 Global laws and the longest run

Theorem 7.4 (Finite-prefix Poisson approximation). *In a fixed logarithmic band, for every $\varepsilon > 0$,*

$$\begin{aligned} d_{\text{TV}} \left(\mathcal{L} \left(\sum_{2 \leq x < M} J_{x,L} \right), \text{Pois}(M 2^{-L}) \right) \\ \ll \min\{1, \lambda_M (e^{-V_M + o(\nu_M)} + M^{-1/3+\varepsilon})\} + e^{-c \log M / \log \log M}. \end{aligned} \quad (7.10)$$

For $W_{M,L} = C_L + \sum_{2 \leq x \leq M-L+1} J_{x,L}$, enlarge the constant multiplying the existing deep-start remainder and add

$$2^{-\pi(L)} + O(L 2^{-L}). \quad (7.11)$$

In particular $\mathbb{P}(R_M < L) = e^{-M 2^{-L}} + o_C(1)$ at critical lengths.

Proof. Retain the single interval $U_{M,1/2}$. The proof of Theorem 4.1 uses only support heights $O(M)$, the one-window mass, the edge count and the separated-pair profile. These are supplied

by (2.4), (4.2) at upper scale M , and Theorem 3.1. Thus (4.8) applies on the retained interval with mean $|U_{M,1/2}|2^{-L}$. Restoring deep starts costs (7.7); the deterministic target-mean discrepancy is $O(\lambda_M M^{-1/2})$. This proves (7.10).

The event $W_{M,L} = 0$ is exactly $R_M < L$: every long run contained in the prefix either begins at one or has a left-maximal start at most $M - L + 1$. The full interior count differs only if the border event occurs or a start lies in the excluded mask $M - L + 2 \leq x < M$. This mask has at most L sites and lies above $2L^2$ for large M . The mask bound in the proof of proposition 7.3 gives total mass

$$L2^{-L} + O(e^{CL/\log L} 2^{-gL}) \leq L2^{-L} + O(e^{-c \log M / \log \log M}).$$

Absorb the second term into the remainder already present in (7.10). The exact border probability then proves (7.11) and the void law. $\square$

Corollary 7.5 (Almost-sure asymmetric envelopes). *For every fixed $\varepsilon > 0$, with probability one, eventually*

$$R_M - \log_2 M \geq -\log_2(\log \log M) - O(1), \quad (7.12)$$

$$R_M - \log_2 M \leq \log_2(\log M) + (1 + \varepsilon) \log_2(\log \log M) + O_\varepsilon(1). \quad (7.13)$$

In particular $R_M = \log_2 M + O(\log \log M)$ almost surely.

Proof. Set $M_k = 2^k$ and, for sufficiently large k , choose

$$L_k^- = \lfloor k - \log_2(\log k) - 1 \rfloor, \quad L_k^+ = \lceil k + \log_2 k + (1 + \varepsilon) \log_2(\log k) \rceil.$$

The lower intensity satisfies $2 \log k \leq M_k 2^{-L_k^-} < 4 \log k$, so the target void probability is at most k^{-2} . The upper intensity is at most $1/[k(\log k)^{1+\varepsilon}]$, a summable sequence. Every error in (7.10)–(7.11) is also summable: the cutoff error is a polynomial in k times $e^{-c\sqrt{k \log k}}$, and the deep-start and border errors are at most $e^{-c'k/\log k}$. Borel–Cantelli gives $L_k^- \leq R_{2^k} < L_k^+$ for all sufficiently large k , almost surely. For $2^k \leq M < 2^{k+1}$ use $R_{2^k} \leq R_M \leq R_{2^{k+1}}$. The change from k to $\log M / \log 2$ and from k to $k + 1$ costs only a bounded additive term. This proves both envelopes. No independence between different scales has been used, and no matching lower bound for the upper fluctuations is asserted. $\square$

Theorem 7.6 (Macroscopic exact-mark transport). *Let $a_M = o(\log M)$ be a nonnegative integer, $b_M = \lfloor \log_2 M \rfloor$, $L_M = b_M - a_M$, and $\lambda = M 2^{-L_M}$. Define*

$$\mathcal{E}_{M,a_M}^{\text{base}} = \sum_{x=2}^{M-L_M+1} \sum_{r \geq -a_M} K_{x,b_M+r}^{\text{ex}} \delta_{(x/M,r)}.$$

Its target has independent Poisson masses 2^{-b_M-r-1} on the same lattice, and total intensity $(M - L_M)2^{-L_M}$. For $A_M \in \mathcal{F}_{Y_M^\#}$ with $\mathbb{P}(A_M) > 0$, put $I_M^{\text{info}} = -\log \mathbb{P}(A_M)$; the full-field distance is bounded by

$$C_\varepsilon e^{I_M^{\text{info}}} \left[\lambda(1 + \lambda)(e^{-V_M + o(\nu_M)} + M^{-1/3+\varepsilon}) + \lambda M^{-1/2} + e^{-c \log M / \log \log M} \right]. \quad (7.14)$$

It tends to zero under the hard labelled budget $I_M^{\text{info}} + \log \lambda + \log(1 + \lambda) \leq V_M - c' \nu_M$. The movable-cutoff labelled budget and the aggregated one-factor budget of Theorem 5.8 also hold, with N replaced by M and exact-level target means $(M - L_M)2^{-b_M-r-1}$.

Proof. On $[\lceil M^{1/2} \rceil, M - L_M + 1]$ use the finite marked comparison with the macroscopic master profile; the identities (5.15) still collapse the separated-pair term to the base threshold event. Restore lower starts with (7.7) and condition using Lemma 6.1. Remove the exact-mark tail using (5.16) and the global scalar bound. A cutoff $E = \lceil (I_M^{\text{info}} + V_M + \log(2 + \lambda)) / \log 2 \rceil$ is $o(\log M)$ whenever the bound is nontrivial; the deep-start tail error is absorbed by the last term of (7.14). This proves the displayed bound. For the movable cutoff repeat with $w = \hat{V}_M$. For aggregation, sum over positions before applying the directional Stein factor; the deep-start error is negligible under its information budget. Section C.6 details the finite-cutoff calculation. $\square$

The superscript “base” is important: only the base L_M -window is required to be contained. The exact right endpoint of a larger marked run may lie beyond M . This is not the list of right-censored maximal runs of the finite sequence. All measurable functionals of this base-contained field, including spatial masks and order statistics, transfer under the labelled comparison. Aggregated couplings do not retain locations.

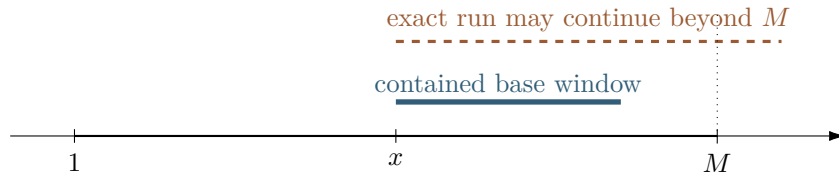


Figure 3: Containment of a base window does not censor an exact mark. The longest-run void event instead uses the threshold-specific upper start $M - L + 1$. The two conventions agree where stated, but are not interchangeable.

7.4 Relative bulk approximation and the crossover

Assume from now on $L \rightarrow \infty$, $L = O(\log M)$ and $\lambda_M = M2^{-L} \rightarrow 0$. Fix $0 < \delta < 1$ and put $H_M = \lceil M^\delta \rceil$, $I_{M,L}^{\text{bulk}} = [H_M, M - L + 1] \cap \mathbb{Z}$ and $\Xi_{M,L}^{\text{bulk}} = \sum_{x \in I_{M,L}^{\text{bulk}}} J_{x,L} \delta_{x/M}$. Let $\Pi_{M,L}^{\text{bulk,lat}}$ have independent Poisson masses $p = 2^{-L}$ at those same sites; its intensity is $\lambda_M^{\text{bulk}} = |I_{M,L}^{\text{bulk}}|p \sim \lambda_M$. Fix an upper-band constant A_0 with $L \leq A_0 \log M$ for all large M . The rare-intensity assumption supplies the positive lower band. Constants and little-oh terms in this subsection may depend on δ , A_0 and the displayed error exponent, but not on M or L .

Theorem 7.7 (Relative signed marked bulk kernel). *Let $\mathcal{K}_{M,L}^{\text{bulk}} = (K_{x,e}^s)_{x \in I_{M,L}^{\text{bulk}}, e \geq 0, s \in \{\pm 1\}}$ and let $\mathcal{P}_{M,L}^{\text{bulk}}$ have independent Poisson coordinates of means 2^{-L-e-2} on this same index set. At $Y = Y_M^\dagger$, for every fixed $0 < \varepsilon < 1/3$,*

$$\mathbb{E} d_{\text{TV}} \left(\mathcal{L}(\mathcal{K}_{M,L}^{\text{bulk}} \mid \mathcal{F}_Y), \mathcal{L}(\mathcal{P}_{M,L}^{\text{bulk}}) \right) \ll_{\delta, A_0, \varepsilon} \lambda_M \left(e^{-V_M + o(\nu_M)} + M^{-1/3+\varepsilon} \right). \quad (7.15)$$

In particular, summing marks and signs gives

$$\mathbb{E} d_{\text{TV}}(\mathcal{L}(\Xi_{M,L}^{\text{bulk}} \mid \mathcal{F}_Y), \mathcal{L}(\Pi_{M,L}^{\text{bulk,lat}})) = o(\lambda_M). \quad (7.16)$$

Proof. For a finite mark cutoff E , repeat the signed comparison of Theorem 5.6 on this single macroscopic interval. With $Q = L + E + 1$ and $p = 2^{-L}$ its error is at most

$$Cp(\mathcal{M}_B + |D_Y(Q)|) + Cp^2(MQ + E_Y(Q) + R_{2,\delta}(M, L)).$$

The signed marks are summed before the separated-pair bound, so the relation sum is at the base length L . Divide by $Mp = \lambda_M$. The defect and support terms become $M^{-1/2+o(1)}$ and

$e^{-V_M+o(\nu_M)}$, respectively. The edge term becomes $\lambda_M(e^{-V_M+o(\nu_M)} + M^{-1+o(1)})$. By (3.4) and $2^B = 2/p$, the relation term is $O_\varepsilon(M^{-1/3+\varepsilon}(1 + \lambda_M))$. The local term pQ is smaller because $\lambda_M \rightarrow 0$.

Take $E = \lceil V_M / \log 2 \rceil$. Then $E = o(\log M)$ and the base and maximal support lengths stay in a fixed logarithmic band. The target tail has mass at most $\lambda_M 2^{-E-1}$. The actual tail is the event that some retained site starts a run at length $L + E + 1$; the macroscopic masked first-moment bound gives the same upper bound times $1 + M^{-1/2+o(1)}$. Couple both tails to zero and average. Since $\lambda_M \leq 1$ for large M , the preceding bounds give (7.15). Measurable contraction proves (7.16). $\square$

Every coordinate of $\mathbf{B}_L$ depends only on prime signs up to $2L^2 + L < Y_M^\#$. The stable product lift therefore gives

$$d_{\text{TV}} \left(\mathcal{L}(\mathbf{B}_L, \Xi_{M,L}^{\text{bulk}}), \mathcal{L}(\mathbf{B}_L) \otimes \mathcal{L}(\Pi_{M,L}^{\text{bulk,lat}}) \right) = o(\lambda_M) = o(q_L + \lambda_M). \quad (7.17)$$

This is asymptotic factorization at the indicated error scale, not exact finite- M independence.

Theorem 7.8 (Two-source rare tail and conditional location). *Under the preceding rare-intensity assumptions,*

$$\mathbb{P}(R_M \geq L) = q_L + \lambda_M + o(q_L + \lambda_M). \quad (7.18)$$

Let $X_{M,L}^*$ be the least start of a contained run of length at least L , on the event that one exists. If $L - \log_2 M - \pi(L) \rightarrow s \in \mathbb{R}$, then

$$\mathcal{L}(X_{M,L}^*/M \mid R_M \geq L) \implies \frac{1}{1+2^{-s}} \delta_0 + \frac{2^{-s}}{1+2^{-s}} \text{Leb}_{[0,1]}. \quad (7.19)$$

The source-resolved version assigns a microscopic start its location divided by L^2 , and a bulk start its location divided by M ; its limit is the same mixture with the two source labels retained. The weights also have their evident degenerate limits when $s \rightarrow \pm\infty$.

Proof. The mesoscopic region has mass $o(q_L + \lambda_M)$ by proposition 7.3. Under the product target in (7.17), a simultaneous microscopic and bulk hit has probability $O(q_L \lambda_M)$, and at least two bulk hits have probability $O(\lambda_M^2)$. Both are negligible on the combined scale. This proves (7.18). Given a microscopic hit, Theorem 7.1 puts the start at one with probability $1 - o(1)$. Given exactly one bulk target point, its location is uniform on the common lattice. Finally

$$\frac{\lambda_M}{q_L} = 2^{-(L - \log_2 M - \pi(L))} (1 + o(1)) \longrightarrow 2^{-s}.$$

Divide the two one-hit masses by (7.18) and pass from the uniform grid to Lebesgue measure weakly. All total-variation comparisons precede this final diffuse limit. $\square$

7.5 The prime clock

Let $p_1 = 2 < p_2 < \dots$. Almost surely $J_\partial = \min\{j : f(p_j) = -1\}$ is finite and the maximal initial positive run has length $\ell_\partial = p_{J_\partial} - 1$: every smaller integer has only positive prime factors, whereas that negative prime changes sign. On this full-measure event and $C_L = 1$, put $G_L = J_\partial - \pi(L) - 1$; assign $G_L = 0$ elsewhere whenever a globally defined random variable is needed. Then

$$\mathbb{P}(G_L = k \mid C_L = 1) = 2^{-(k+1)}, \quad \mathbb{P}(\ell_\partial - L \geq h \mid C_L = 1) = 2^{-[\pi(L+h) - \pi(L)]}. \quad (7.20)$$

Thus prime rank, not integer distance, is its geometric clock. The boundary theorem makes the law of G_L given $B_L^{\text{mic}} > 0$ $O(2^{-\delta_0 \pi(L)})$ -close to ν_{geo} . The prime number theorem gives $\ell_{\partial}/L \rightarrow 1$ conditionally, but the unscaled integer overshoot is not tight: for $L = m! + 1$ the following $m - 1$ integers contain no prime.

7.6 A lattice crossover with two geometric clocks

The two branches have the same geometric law but different clocks: additional integers in a bulk run, and additional positive prime signs at the initial boundary. A moving discrete target states their joint law without requiring a limiting ratio of the two rare masses.

Write $\alpha_L = 2^{-\pi(L)}$ and $\lambda^{\text{bulk}} = \lambda_M^{\text{bulk}}$. On $\{R_M \geq L\}$, let $X^* = X_{M,L}^*$ and define

$$\Gamma_{M,L} = \begin{cases} (\partial, 1, +1, G_L), & X^* = 1, \\ (\text{bulk}, X^*, f(X^*), \ell_{X^*} - L), & X^* \in I_{M,L}^{\text{bulk}}, \\ \dagger, & \text{otherwise.} \end{cases}$$

Here $\dagger$ records an exceptional start in the intervening region. Let $\nu_{\pm} = (\delta_{-1} + \delta_{+1})/2$, and let $\text{Unif}(I_{M,L}^{\text{bulk}})$ be the uniform law on that integer grid.

Theorem 7.9 (Sign-resolved, two-clock crossover). *Under the rare-intensity assumptions of Theorem 7.8,*

$$d_{\text{TV}}(\mathcal{L}(\Gamma_{M,L} \mid R_M \geq L), Q_{M,L}) \longrightarrow 0, \quad (7.21)$$

where the deterministic probability law on the same discrete state space is

$$\begin{aligned} Q_{M,L} &= \frac{\alpha_L}{\alpha_L + \lambda^{\text{bulk}}} \delta_{(\partial, 1, +1)} \otimes \nu_{\text{geo}} \\ &\quad + \frac{\lambda^{\text{bulk}}}{\alpha_L + \lambda^{\text{bulk}}} \delta_{\text{bulk}} \otimes \text{Unif}(I_{M,L}^{\text{bulk}}) \otimes \nu_{\pm} \otimes \nu_{\text{geo}}. \end{aligned} \quad (7.22)$$

No convergence of $\lambda^{\text{bulk}}/\alpha_L$ is needed. In particular,

$$\mathbb{P}(f(X^*) = +1 \mid R_M \geq L) = \frac{\alpha_L + \lambda^{\text{bulk}}/2}{\alpha_L + \lambda^{\text{bulk}}} + o(1). \quad (7.23)$$

The bulk excess in (7.21) may instead be measured using the right-censored run length in the finite word $(f(1), \dots, f(M))$.

Proof. Fix $K \geq 1$. Count the consecutive positive signs among the first K primes after L , capped at K , and set the result to zero off $C_L = 1$. This is a genuine finite-cylinder random variable, agreeing almost surely with $G_L \wedge K$. For fixed K , Bertrand's postulate places these primes below $2^K L < Y_M^{\#}$ for large M . The stable lift applied to Theorem 7.7 therefore retains the microscopic vector and this clock jointly with *all* signed bulk marks, at error $o(\lambda_M)$. No bulk-mark truncation is needed here. The interior microscopic and mesoscopic errors are $o(\alpha_L + \lambda_M)$ by Theorem 7.1 and proposition 7.3.

Write $\alpha = \alpha_L$ and $b = \lambda^{\text{bulk}}$. In the product target, keep either a boundary hit with no bulk point, or no boundary hit with exactly one bulk point. This event has exact mass

$$e^{-b}[\alpha + (1 - \alpha)b].$$

Its conditional source weights are $\alpha/[\alpha + (1 - \alpha)b]$ and $(1 - \alpha)b/[\alpha + (1 - \alpha)b]$. Their mixture differs from the weights in (7.22) by at most

$$\frac{\alpha^2 b}{(\alpha + b)[\alpha + (1 - \alpha)b]} \leq b.$$

The discarded simultaneous and multiple hits cost $O(ab + b^2) = o(\alpha + b)$. Given the one bulk point, its site, sign and excess have the independent laws in (7.22). Normalize all comparison errors by $\alpha + b$, not by the smaller source mass. This proves the comparison with the boundary clock capped at K , uniformly in the moving weights.

Under the actual law, a recorded boundary tail $G_L > K$ has conditional probability at most 2^{-K-1} , since $\mathbb{P}(C_L = 1, G_L > K) = \alpha 2^{-K-1}$ and $\mathbb{P}(R_M \geq L) \geq \alpha$. The target boundary tail is also at most 2^{-K-1} . Let $M \rightarrow \infty$ first and then $K \rightarrow \infty$. This proves (7.21); projecting the sign gives (7.23).

For the last assertion, under the one-hit bulk target a mark is censored at site x only if $e > M - x + 1 - L$. The average of this geometric tail is

$$\frac{1}{|I_{M,L}^{\text{bulk}}|} \sum_{x \in I_{M,L}^{\text{bulk}}} 2^{-(M-x+2-L)} \leq \frac{1}{|I_{M,L}^{\text{bulk}}|} = o(1).$$

The total-variation comparison transfers this estimate, so replacing that mark changes the conditional law by $o(1)$. $\square$

After scaling sites by M , convergence of the ratio recovers the mixture in Theorem 7.8, now with signs and both overshoot clocks retained. The spatial limit is weak; the stronger statement (7.21) is entirely on the prelimit lattice. Its $o(1)$ error is on the combined source scale. It does not supply a relative estimate for the smaller source when its mixture weight tends to zero.

7.7 Affine information at the boundary

Theorem 7.10 (Affine-conditioned crossover). *Let $A_M = \{G_M X(Y_M^\sharp) = b_M^{\text{aff}}\}$ be a compatible affine cylinder of rank r_M , and assume compatibility also with $C_L = 1$. Set*

$$U_L = \text{span}\{e_p : p \leq L\}, \quad \kappa_M = \dim(\text{row } G_M \cap U_L), \quad d_M = \pi(L) - \kappa_M.$$

Whenever $L \leq Y_M^\sharp$, the finite identity $\mathbb{P}(C_L = 1 \mid A_M) = 2^{-d_M}$ holds exactly. For the asymptotic conclusions, let $M, L \rightarrow \infty$ with $L = O(\log M)$ and $\lambda_M = M 2^{-L} \rightarrow 0$. Fix $c > 0$ and assume $r_M \log 2 \leq V_M - c\nu_M$ eventually. The affine equations, their number and their right-hand sides may all vary along the sequence. Then the rare-tail formula is

$$\mathbb{P}(R_M \geq L \mid A_M) = 2^{-d_M} + \lambda_M + o(2^{-d_M} + \lambda_M).$$

If $L - \log_2 M - d_M \rightarrow s$, the conditional location law given A_M and $R_M \geq L$ is (7.19) with the same weights. Microscopic non-vacancy under A_M is concentrated on the border.

Proof. The stacked cylinder and border system has rank $r_M + \pi(L) - \kappa_M$, proving the exact mass ratio. Write $I = r_M \log 2$. Apply the factor e^I to the quantitative bound (7.15) before taking a limit. Its normalized cutoff term is at most $e^{-c\nu_M + o(\nu_M)}$, and its polynomial term vanishes because $I = O(V_M) = o(\log M)$. Hence the conditional bulk error is $o(\lambda_M)$. The microscopic interior mass, after conditioning and division by 2^{-d_M} , is at most $O(e^I 2^{-\delta_0 \pi(L) - \kappa_M}) = o(1)$, since $I = O(V_M) = o(\pi(L))$. Similarly the two terms of (7.8) become $o(\lambda_M)$ and $o(2^{-d_M})$. The

conditional stable lift and the exact one-hit normalization in the proof of Theorem 7.9 now apply, with total normalization $2^{-d_M} + \lambda_M^{\text{bulk}}$. $\square$

The parameter κ_M counts linear consequences supported wholly on the border coordinates, not input equations merely mentioning them. No condition on future primes is needed for the rare tail, microscopic concentration or location laws. The geometric future mark requires a separate condition. Let $q_1 < q_2 < \dots$ be the primes after L . If, for each fixed K , eventually in M with a threshold that may depend on K ,

$$(\text{row } G_M + U_L) \cap \text{span}\{e_{q_1}, \dots, e_{q_K}\} = \{0\},$$

then the geometric boundary mark and the full mixture of Theorem 7.9 survive under A_M , with α_L replaced by 2^{-d_M} . First fix K and pass to the size limit, then let $K \rightarrow \infty$; no common threshold for all K or neutrality for a growing $K(M)$ is assumed. In general, if the extended affine stack is compatible, $\mathbb{P}(G_L \geq K \mid A_M, C_L = 1)$ equals $2^{-[K - \dim((\text{row } G_M + U_L) \cap \text{span}\{e_{q_1}, \dots, e_{q_K}\})]}$; if incompatible, the tail is zero. This separates the information that changes the border mass from information that changes its future mark. [Section E.7](#) records the rank calculation.

8 What the method isolates

Prime signs are independent, but their products satisfy exact identities at arbitrarily large distances. The proof uses both facts. It counts the homogeneous relation spaces created by multiplication, then conditions on the prime environment that makes a dependency graph available. The three-term profile (3.3), and its full-value extension (3.25), are the arithmetic outputs. The independent Poisson field is obtained only after the separate marginal, local-overlap and prime-support estimates have been verified.

8.1 A class of observables, rather than one count

The dictionary theorem separates local word combinatorics from nonlocal arithmetic. Changing the prescribed signs does not change a valuation matrix. Prescribing their absolute orientation removes at most two parity conditions in a pair of windows; the host count controls the additional relations. Consequently the same profile handles a growing dictionary and retains the identity and position of every rare word. Its overlap condition holds for most dictionaries in the stated regime, and explicit marker families make it zero. Short suffix–prefix matches can instead create local clusters independently of any multiplicative relation; such dictionaries require a separate analysis.

Long runs provide two complementary examples. Counting left-maximal starts gives a Poisson field with geometric excesses and fair signs. Counting all constant windows retains the clusters, and the field contracts to the explicit compound-Poisson law of Corollary 5.7. Both statements come from the same comparison, not from two unrelated moment arguments. The asymmetric almost-sure bounds in Corollary 7.5 use its summable quantitative errors; they do not require independence across scales.

The boundary is a different arithmetic regime. Its initial positive cylinder imposes $\pi(L)$ independent prime constraints, whereas other microscopic starts have an additional rank cost. The competition between $2^{-\pi(L)}$ and $M2^{-L}$ therefore produces a change in the location and sign of a rare long run. The two-clock theorem retains more than the limiting source weights: it gives a prelimit lattice law, with prime-rank overshoot at the boundary and integer overshoot in the bulk. The logistic formula is a consequence of normalizing these two masses. The substantive

inputs are the microscopic rank surplus, the relative marked bulk kernel and the negligible region between them.

8.2 Arithmetic and probabilistic limits of the estimates

Three terms reach $N^{5/3}$ at criticality in the present upper bound. Improving only the terminal kernel energy would therefore not improve the total exponent. The rational families in (3.22)–(3.23) provide lower exponents $4/3$ and $3/2$. Sharpening this gap is an arithmetic question. It is distinct from improving the leading subexponential error, which is governed by exceptional supports and shared prime coordinates. For growing dictionaries the relevant quantity is not the full relation sum, but its capped version. Mutual exclusivity at one site gives a probability ceiling, and proposition 3.25 preserves it through the terminal count. This removes the terminal restriction on dictionary size without improving the uncapped $5/3$ exponent. The remaining term $NQ_B^{2/3}$ gives the sufficient exponent $1/2$ in (5.3). A further improvement would need to address this nonterminal contribution or use more information about the dictionary; it does not follow from another terminal refinement alone.

A similar economy occurs in the Diophantine step. All component variables already lie in bounded polynomial height. Localizing their squarefree parts to the primes in the coefficient and the root differences reduces the required count to two factors and a Pell equation. A height-free bound for all integral points would be stronger, but is not needed. The proof uses the hypotheses its application actually supplies.

The scalar, aggregated and spatially labelled comparisons retain different information and therefore incur different costs; the table in section 6 records their sufficient budgets. The local resolution condition is separate: the absolute error must be negligible relative to the probability of the event being resolved.

8.3 Further questions

A natural extension is a dictionary with substantial short overlaps. The arithmetic profile still bounds distant interactions, but a new cluster analysis is needed to transfer the local clump structure to this multiplicative model. Such compound-Poisson descriptions for Markov word families are developed in [28, 30]; the problem here is their compatibility with the nonlocal arithmetic dependence. The constant-window compound-Poisson law gives a model case; it is not a classification of periodic or interacting patterns. A second question is the full one-factor growing-intensity range for the spatially labelled field. This would require a sharper treatment of dependencies between several sites, beyond the process comparison used here. Neither question is an assumption of any theorem in this article.

Finally, the run lengths and their phase-indexed extremes remain on an integer lattice. The separate process article [27] develops records, occupation laws and scale flows from the signed field. Its intrinsic integer observables are distinct from its continuous-time embeddings and auxiliary phase removal, which use additional randomization. None supplies an exact continuous finite- N Markov property for the multiplicative sequence.

Formalization and availability

The mathematical proofs are given in this article and its technical companion, with the literature inputs identified where they are used. An associated Lean development is available at [Vulkin-prog/paper-c-lean](#). It is separate from the written proof and is not used as an input to it. No formal certification of the present article or companion is claimed. Checking a Lean proof under external hypotheses does not establish the truth of those hypotheses.

The article and technical companion form one research object associated with Zenodo concept DOI [10.5281/zenodo.21736676](#) and Cambridge Open Engage record [10.33774/coe-2026-z3l74](#). The source archive contains the LaTeX files, compilation instructions, finite diagnostic programs and a file manifest. No version-specific DOI is assigned here. Finite diagnostics detect algebraic and implementation errors; they do not replace the proofs of the asymptotic results.

Declaration on artificial-intelligence assistance

The research was organized and directed by the author with substantial assistance from language-model systems, including Claude Fable 5 and Claude Opus 4.8 (Anthropic), and GPT-5.6 Sol and GPT-6 Astra (OpenAI). These systems were used to explore candidate lemmas and proof routes, draft and reorganize manuscripts, assist with the separate Lean development, write diagnostic programs, check references and build artifacts, and conduct adversarial reviews through separate instances.

The author determines the claims and scope and accepts responsibility for the arguments, references and any remaining errors. AI-assisted checks are not independent human peer review. The mathematical claims rest on the printed arguments and the stated external results, not on model output or the success of finite diagnostic tests.

Competing interests

The author declares no competing interests.

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